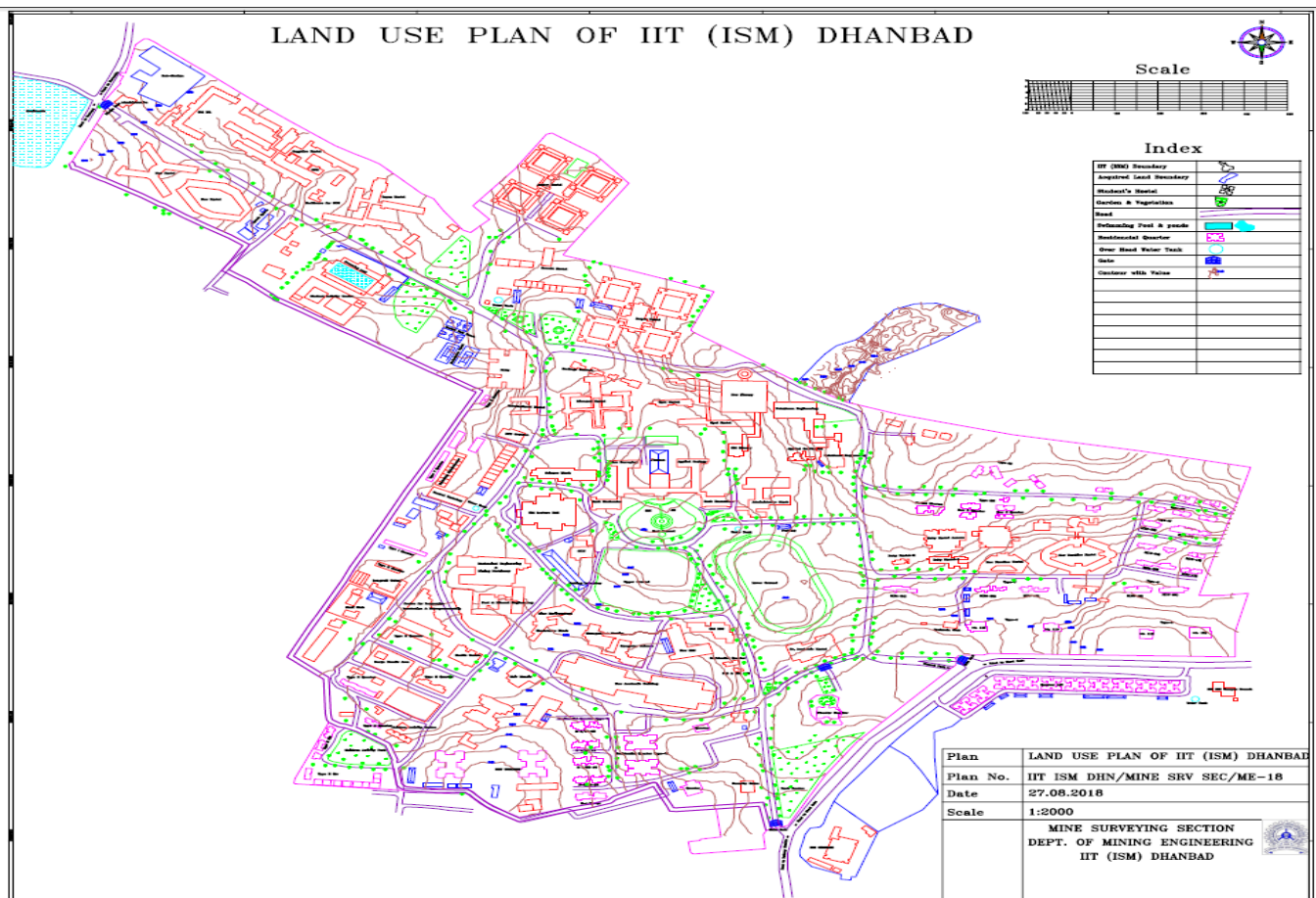


LABORATORY MANUAL
III SEMESTER B.TECH MINING ENGINEERING
Course Name: Mining Surveying Laboratory. Course Code: MNC 203



Lab In-charge: Prof. Dheeraj Kumar
Assistant Lab In-charges: Prof. V. V. Govind Kumar
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Technical Support: Shri Sunil Kumar
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Department of Mining Engineering

Mine Surveying Section

Indian Institute of Technology (Indian School of Mines), Dhanbad

LIST OF SURVEYING PRACTICALS

Course: III Sem. B. Tech (Mining Engineering)

Semester: Monsoon. Course Code: MNC 203

Sl. No .	Aim of experiment	No. of Practical	Learning Outcome
1	Study and sketch of linear measuring instruments (Tapes and Distometer) and carryout linear measurement.	1	Measurement of linear distances using direct and indirect methods
2	Introduction to various angular measurement instruments and measurement of angles using theodolite	1	Measurement of horizontal and vertical angles
3	Closed Traversing using theodolite and distribution of closing errors	1	Transferring coordinates from known baseline
4	Introduction to various Levelling instruments and carryout profile leveling using auto level	1	Transferring of RLs from known benchmark
5	Study and sketch of Total Station and determination of coordinates using Total Station	2	Usage of Total Station and its application in data acquisition
6	Preparation of digital mine plans using Total station data	1	Processing of Total station data and Preparing digital plans
7	Study and sketch of electronic gyro- theodolite (Gyromat-3000) and determination of Gyro-north	1	Obtaining Gyro-meridian and gyro bearing of a survey line using digital gyro theodolite
8	Study and sketch of Nadir Plummet and LASER and demonstration for mine correlation survey	1	Usages and procedure of the modern way of mine correlation survey.
9	Study of GNSS system and carryout GNSS survey in relative point positioning.	1	DGPS surveying for delineation of surface features
10	Study of Terrestrial LIDAR (TLS) and demonstration for Topographical Surveying	1	Application of TLS for surface terrain mapping and modelling

PATTERN OF ASSESSMENT

<i>S.no.</i>	<i>Head</i>	<i>Distribution of marks</i>	
1.	<i>Practical notebook</i>	20	<i>Sessional</i>
2.	<i>Field book</i>	10	
3.	<i>Punctuality & Performance</i>	20	
4.	<i>Lab exam</i>	30	<i>End -term</i>
5.	<i>Viva-voce</i>	20	
<i>Total</i>		100	

GENERAL INSTRUCTIONS OF THE LABORATORY

1. LABORATORY SAFETY GUIDELINES:

- *Every student must be with required PPEs during the lab experiment.*
- *Ensure that all safety devices are adequate, appropriate and good working condition.*
- *Every Student should follow the SOP (Stander Operating Procedures) of all instruments.*
- *Every Student should know the location of First AID KIT (s) and Fire extinguisher.*

2. RULES & GUIDELINES FOR LABORATORY EXPERIMENT:

- *All students will take a sit for their attendance and understand about the laboratory experiment.*
- *Before starting the Laboratory experiment, follow all written and verbal instructions carefully.*
- *Before use any instrument or software, read and follow the SOP carefully.*
- *Any difficulties, error, missing of any parts of the instruments or break-down, must be reported to the teacher, instructor or Laboratory superintendent.*

3. DO'S & DON'TS DURING LABORATORY EXPERIMENT:

- *Conduct yourself in a responsible manner at all times in the laboratory.*
- *Proper dress should maintain during the laboratory class.*
- *Observe good housekeeping practices.*
- *Issue and hand over the instruments and accessories from proper place.*
- *Students are not allowed to work in the Laboratory alone.*
- *Don't wander around the room, distract other students, or interfere with others.*
- *Do not enter with food and drink beverages in the laboratory.*
- *Don't touch any instrument, accessories and computer without permission.*

EXPERIMENT: 1

Name of the Experiment:

Study and sketch of linear measuring instruments (Tapes and Distometer) and carryout linear measurement.

Objective:

- 1. To take measurements to determine the relative positions of the existing features on or near the ground.*
- 2. To layout or to mark the positions of the proposed structures on the ground.*
- 3. To determine areas, volumes and other related quantities.*

Linear Measurement:

Linear measurement is a measurement of length. There are three methods of linear measurements. Such as

- Direct Method*
- In-Direct Method*
- EDM Method*

Approximate methods of linear measurement are

- Pacing*
- Passometer*
- Pedometer*
- Odometer*
- Speedometer*
- Measuring Wheel*

Instruments used in Chain Surveying:

- Chains*
- Tapes*
- Arrows*
- Ranging Rod & Offset Rod*
- Pegs*
- Plumb-bob*
- Optical Square*
- Spring Balance*

Chain:

The chain consists of a number of large links made of galvanized mild steel wire of 4 mm diameter. Each end of the large links is bent into a loop. These links are connected in series by small connecting links. The connecting links consist of three rings; the outer two rings are of oval-shape where the central ring is circular. These connecting links provide flexibility to the chain. The two ends of the chain have brass handles which are used for pulling and dragging the chain. The handles have swivel joints to provide flexibility so that it is possible to swivel the handle round the bolt without its twisting. The length of the links is measured from the centre of the connecting central ring to the centre of the next connecting central ring, and it is 200 mm for all links except the end links. End links also 200 mm length including the brass handle. The surveying chains are of four different lengths according to IS: 1492-1970, (1) 5 m (25 links), (2) 10 m (50 links), (3) 20 m (100 links) and (4) 30 m (150 links). To enable the reading of fraction of a chain length without difficulty, brass tallies or tags and rings are attached to the chain at some fixed interval. Every metre length of the chain should be accurate to within ± 2 mm under a tension of 80 N as per IS specification. The overall length of the chain should be within the limits 30 ± 8 mm for 30 m chain.

Types of Chains:

- Metric chain
- Gunter's chain or Surveyor's chain
- Engineer's chain
- Revenue chain
- Steel band or Band chain

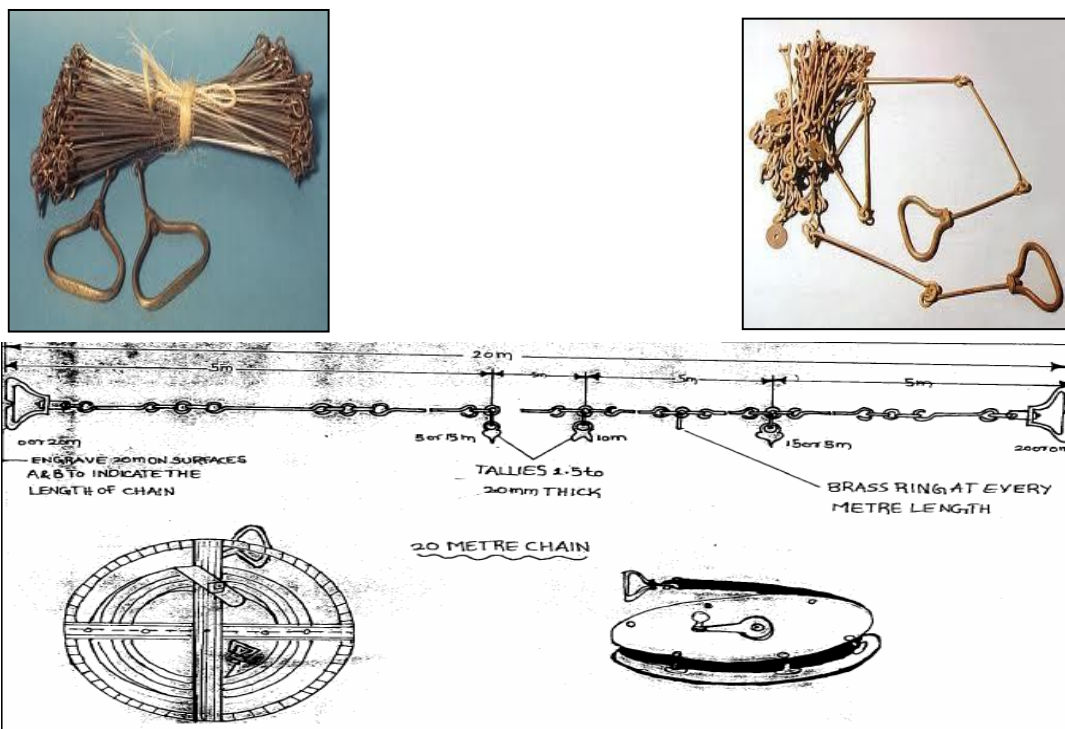


Fig. 1

Metric chains are made in lengths 20m and 30m. Tallies are fixed at every five-meter length and brass rings are provided at every meter length except where tallies are attached.

Tapes:

Measuring Tapes can be classified into 5 types, depending upon the material used in their manufacture.

- Linen or cloth tapes
- Glass- fibre tapes
- Metallic tapes
- Steel tapes
- Invar tapes

Linen or cloth tapes:

These tapes are made of linen or cloth. The linen tape consists of a long strip, 10 mm to 15 mm wide. These tapes are available in lengths of 10 m, 20 m, 25 m, 30 m. The tape is light and handy but not very accurate. The linen tape changes in length due to shrinkage and stretching. It is also liable to twist and tangle. The tape is also affected by dampness and heat. The plastic – coated tapes are tougher than ordinary linen tapes and are more durable. These are relatively more water proof.



Fig. 1.4 Linen or cloth tape

Glass – fibre tapes:

These tapes are similar to linen and plastic coated tapes but these are made of glass- fibre. The tapes are quite flexible, strong and non-conductive. These can be

used even in the vicinity of electrical equipment. These tapes do not stretch or shrink due to changes in temperature or moisture. These tapes are available in various lengths. These tapes are more useful than linen tapes.



Fig. 1.5 Glass-fiber tape

Metallic Tapes:

The metallic tapes are similar to linen tapes but are made of water proof fabric or glass fibre in which metallic wires are interwoven. Generally, the strands of cooper wire are used for interweaving, but brass or bronze wires are also sometimes used. The metallic wires prevent excess stretching of the tape. The metallic tapes are made in width of 10 mm to 15 mm.



Fig. 1.6 Metallic tape

Steel tapes:

The steel tapes are more accurate than metallic tapes. The tapes are made of steel or stainless steel strip. The tapes of widths 6 mm, 9.5 mm, 13 mm and 16.0 mm are commonly used. The thickness of the tap used is 0.15 mm, 0.20 mm or 0.4 mm. The normal length of the tapes is 1 m, 2 m, 10 m, 15 m, 20 m, 30 m, or 50 m. The strips of tapes of 1 m and 2m may be flat or curved.



Fig. 1.7 Steel tape

Invar tapes:

Invar tapes are made of an alloy of steel (64%) and nickel (36%); which has a very low coefficient of thermal expansion. These tapes are generally 6 mm wide, and are available in lengths of 20 m., 30 m and 60 m. The invar alloy is a soft material, and, therefore, the

invar tapes should be handled carefully to avoid bends and kinks. Invar tapes are kept on reels of large diameter to avoid kinking.



Fig. 1.8 Steel tape

Arrows:

Arrows or chain pins are used to mark the position of the ends of the chain on the ground. The arrows are made of good quality, hardened and tempered steel wire of 4 mm diameter and of a minimum tensile strength of 700 70 kg f/mm. The wire is black enamelled with an enamel of uniform thickness. The length of the arrow varies from 25 to 50 cm. As per IS: 1842-1961, the overall length is 400 mm, and the diameter of the loop is 50 mm.

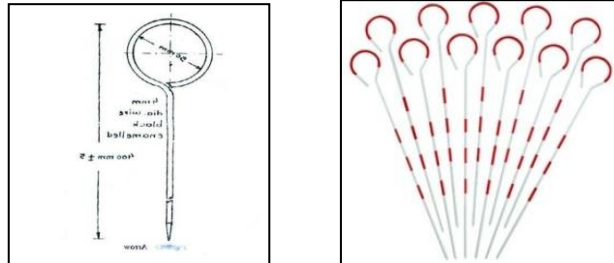


Fig. 1.9 Arrows

Ranging rods:

The process of locating a number of points on a long survey line is called ranging out or ranging. Ranging rods are used to locate intermediate points such that these points lie on the straight line joining the end stations. Ranging is essential before starting the measurement of a line whose length is more than a chain length so that measurement is made along a straight line.

The ranging rods of octagonal cross section are also used. IS: 2288-1963 specifies ranging rods of two lengths. The smaller one is of 2 m length, with 10 bands of 200 mm each. The larger one is of 3 m length, with 15 bands of 200 mm each.

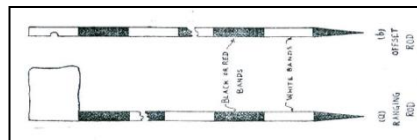


Fig. 1.10 Ranging rods

Pegs:

Wooden pegs are used to mark the position of the survey stations or the end points of a survey line. The pegs are used to mark the intersection of lines and other such points which are more or less permanent, and are required to be retained for some time. The

pegs most commonly used are 25 mm x 25 mm in cross section and 150 mm long. For soft grounds, the pegs of 40 mm length are used.



Fig 1.11 Pegs

Plumb Bobs:

The plumb bob plays a very important role in surveying. As a freely suspended plumb bob always points towards the gravity, it indicates the direction of the vertical line. A plumb bob is a ball made of brass, of the shape of a pear. It has a fine, hardened steel point. There is an eye or a hook at the top for attaching a string of braided nylon. The length of the cord is about 2 m and the weight of the plumb generally varies from 2 to 5 N. the length of the bob is about 50 mm.



Fig. 1.12 Plumb bobs

Spring Balance:

A spring balance is a type of weighing scale. It consists of a spring fixed at one end with a hook to attach an object at the other. It works by Hooke's Law, which states that the force needed to extend a spring is proportional to the distance that spring is extended from its rest position. Therefore, the scale markings on the spring balance are equally spaced. A spring scale only weighs. A spring balance can be calibrated for the accurate measurement of mass in the location in which they are used. A spring scale will only read correctly in a frame of reference where the acceleration in the spring axis is constant. The scale on top would read slightly heavier due to also supporting the weight of the lower scale itself. Spring balances come in different sizes. A spring balance may be labeled in both units of force (pounds, newtons) and mass (grams, kilograms).



Fig. 1.13 Spring balance

Optical Square:

An optical square is a small pocket instrument. It is used for setting out right angles. It consists of a small circular box about 5 to 65 mm diameter and about 12.5 mm deep. There are three small openings of about 1.5 mm diameter. There are two mirrors inclined at 45° to each other. The horizon glass is fixed in position, wherever the index glass can be adjusted. The optical square is based on the optical principle that if a ray of light undergoes two successive reflections in a plane in right angles to each of the two plane mirrors.

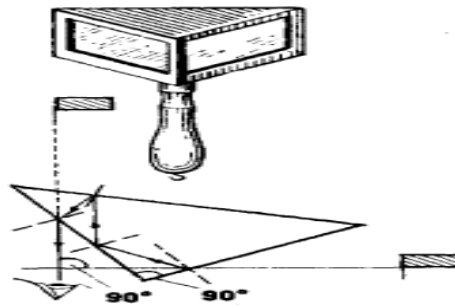


Fig. 1.14 Optical Square

Chain Surveying:

Chain surveying is the method of land surveying in which only linear measurement are made. No angular measurement is taken. Chain surveying is used for areas of small extent on open ground having few simple details. This is the simplest type of surveying and consists of measuring the lengths of the lines marked out in the field. Chain surveying is used for securing data for exact description and marking of the boundaries of a piece of land or for preparing the maps of the area to show various details.

Principles of Chain Surveying:

A triangle is the simple figure which can be plotted from the lengths of the three sides even if the angles are not known. In chain surveying, the area to be surveyed is divided into a framework consisting of triangles. The simplest possible chain surveying consists of only one triangle.

Field work in Chain Surveying: The field work of chain surveying can be divided into the following four stages:

- **Reconnaissance**
- **Marking Stations**
- **Running survey lines**
- **Taking offsets.**

Reconnaissance:

Reconnaissance is the preliminary inspection of the area to be surveyed to have some idea of the terrain and the principal features of the ground. During reconnaissance, the surveyor generally makes an index sketch (or the key plan) to show the principal features, such as building, roads, nallahs, boundaries.

Marking Stations:

After completing reconnaissance, the main stations are marked on the ground so that the stations can be readily located afterwards, if required to the survey is of a temporary character, the stations are generally marked by fixing a ranging rod at each stations.

Running survey Lines:

Survey lines are run to measure the distance between main stations and to locate the adjacent details by offsets. The chaining is usually commenced from the base line and continued to other survey lines.

Taking Offsets:

Offsets are lateral distances measured from the survey line to the detail. The offsets are of two types:

- *Perpendicular offsets or simply offsets,*
- *Oblique offsets or ties.*

Recording Measurements:

The measurement taken in chain surveying is recorded in the field book. The field notes should be

- *Accurate*
- *In a proper format*
- *Complete*
- *Clear and neat*
- *Legible and*
- *Unbiased*

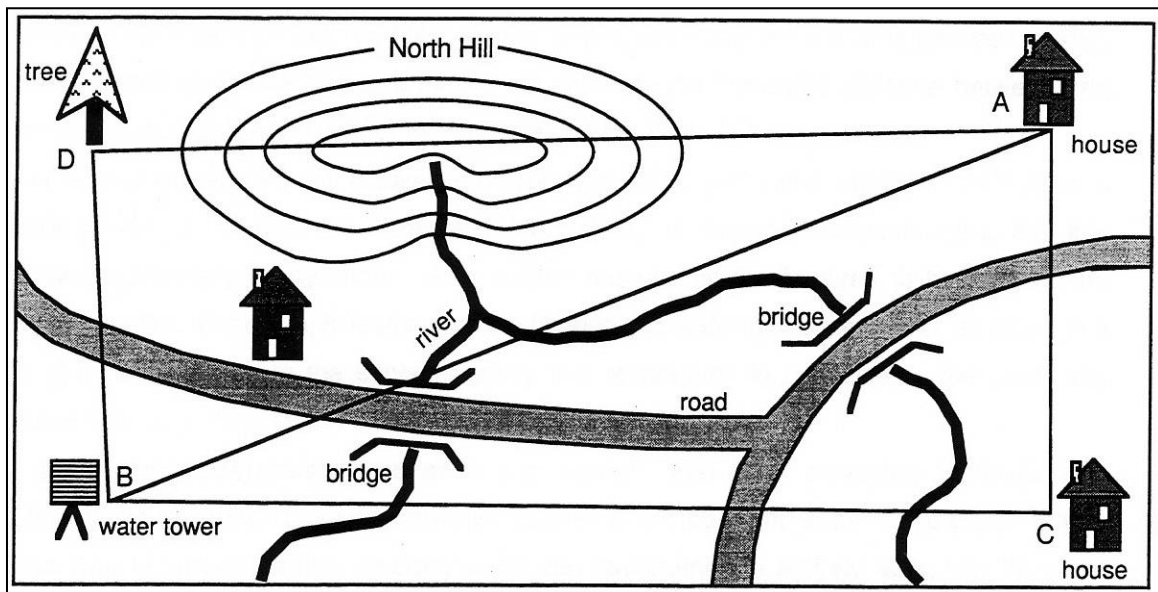


Fig. 1.15 Chain surveying

EXPERIMENT: 2

Name of the Experiment:

Introduction to various angular measurement instruments and measurement of angles using Theodolite.

Vernier Theodolites:

A theodolite is an important used for measuring horizontal and vertical angles in surveying. In the vernier are used for taking the readings. These theodolites are most commonly used in general work. Most of the vernier theodolites can be read angles up to 20''.

Main parts of a vernier theodolite:

The discussion is generally and applies to most of the vernier theodolites manufactured in India and abroad.

- Telescope.
- Trunnion axis.
- Vernier frame.
- Vertical circle.
- Plate level.
- Standard.
- Upper plate.
- Lower plate.
- Inner axis.
- Outer spindle.
- Inner spindle.
- Altitude bubble.
- Levelling screw.
- Clip screw.
- Hook.
- Tribrach.
- Vertical clamp screw.
- Vertical tangent screw.
- Trivet.
- Tripod.
- Tripod leg.

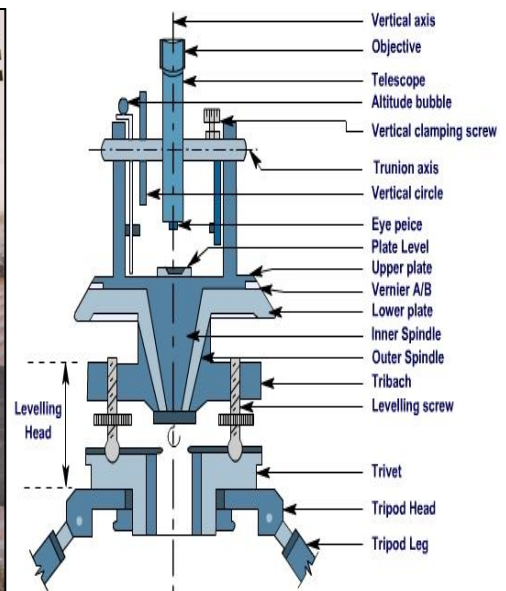


Figure 20.4 Sectional view of a Theodolite

Fig. 2.1 Main parts of vernier theodolite

Telescope:

The telescope is mounted on a horizontal spindle called the horizontal axis or the trunnion axis. The aperture of the objective is generally 38mm. The magnification varies from 18 to 28 diameters. The length of the telescope tube is usually between 100 to 175 mm.

Vertical Circle:

The vertical circle is graduated in degrees, with graduations at 20'. The graduations in each quadrant are numbered from 0 to 90 degree in opposite directions from the two zeroes placed at the horizontal diameter of the circle.

Vernier Frame:

The Vernier frame, also called T-frame or index frame, consists of a vertical leg called clipping arm and a horizontal bar called the index arm.

Vertical Clamp Screw:

The vertical circle and hence the telescope can be clamped at any desired vertical angle by means of the vertical clamp screw.

Altitude Bubble:

A sensitive level tube, called altitude bubble, is attached to the top of the T-frame. The sensitivity of the altitude bubble is usually $20''$ per 2 mm.

Striding Level:

A very sensitive level is sometimes mounted at right angles to the telescope axis. It is used in astronomical observation for levelling the horizontal axis. It is not in theodolite.

Upper Plate:

The upper plate, also called the Vernier plate, supports the standard at its upper surface.

Lower Plate:

The lower plate is also known as the main scale plate or the horizontal circle. The lower plate is graduated in degrees from 0° to 360° with graduations at 20° . The diameter of the horizontal circle is between 100 to 130 mm. This diameter is used to designate the size of the instrument.

Plate Level:

A level tube, called plate level, is mounted on the upper plate. The sensitivity of the plate level is usually $35''$ per 2mm.

Levelling head:

The levelling head consists of two parallel plates, separated by three levelling screws. The upper parallel plate of the levelling head is called the tribrach and the lower one is called the trivet stage or foot plate.

Shifting Head:

The shifting head is a centring device placed below the lower plate but above the tribranch so that the centring may be done after the instrument has been levelled. It consists of two plates which can move relative to each other. When the device is untightened, the whole upper part of the instrument can be slightly shifted horizontally up to about 10 mm diameter.

Magnetic Compass:

The magnetic compass is generally of circular type. It is mounted on the upper plate between the standards. The letter N of the compass is under the objective end of the telescope when the telescope is in the normal position, and the letter S is under the eyepiece end.

Finder- collimator device:

The telescope is generally fitted with a pair of external sights or finder collimator for rough pointing of the telescope towards the object.

Tripod:

Theodolite is mounted on a strong tripod when being used in the field.

Fundamental Lines: There are 5 fundamental or axis of a theodolite.

- Vertical axis.
- Horizontal axis.
- Line of collimation.
- Altitude level axis.
- Plate level axis.

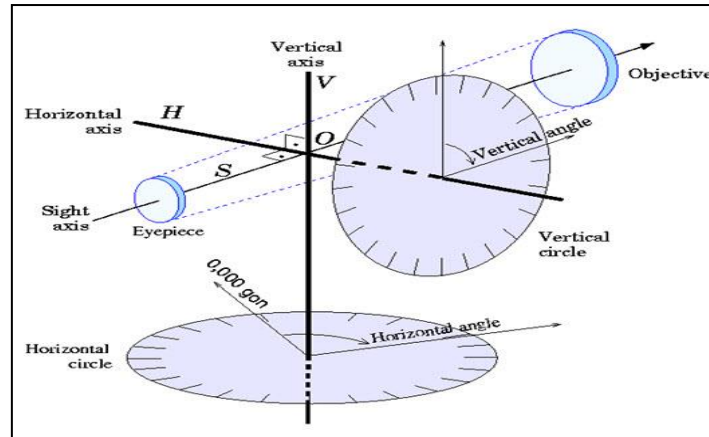


Fig. 2.2 Fundamental axis of theodolite

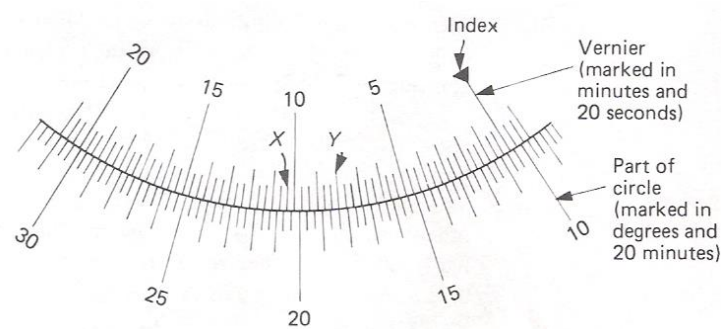


Fig. 2.3 Reading system of Vernier theodolite

When the theodolite is in proper adjustment, the following four conditions between fundamental lines are satisfied.

- The axis of the plate level is perpendicular to the vertical axis.
- The horizontal axis is perpendicular to the vertical axis.
- The line of collimation is at right angles to the horizontal axis.
- The axis of the altitude level is parallel to the line of collimation when it is horizontal, and the vertical reads zero.

The horizontal axis is perpendicular to the vertical axis. It is along the trunnion axis and the line of collimation is visualised as a line along the axis of the telescope. It always remains in a plane perpendicular to the horizontal axis.

The altitude level axis and the plate level axis are perpendicular to the vertical axis.

Functions of clamp and tangent screws:

Clamp Screws:

The beginner finds it difficult to manipulate the clamp screws provided for the upper plate and lower plate. The following three conditions should be clearly understood.

- When the upper clamp screw is tightened but the lower clamp screw is loose, the instrument rotates on its outer axis, without any relative motion between the two plates. The Vernier reading does not change.
- When the lower clamp screw is tightened, but the upper m clamp screw is loose, the instrument rotates on the inner axis with a relative motion between the two plates. The Vernier reading change.
- When both the upper clamp screw and the lower clamp screw are tightened, the instrument cannot rotate at all.

Tangent Screws:

For small movement of plates, corresponding tangent screws are used. Before using any tangent screw, the corresponding clamp screw must be tightened first; it will not work.

- When the upper clamp screw has been tightened, small movement of the upper plate can be made by the upper tangent screw.
- When the lower clamp screw has been tightened, small movement of the lower plate can be made by the tangent screw.

Temporary Adjustments of a Theodolite:

The following five temporary adjustments are required.

Setting up:

For setting up the instrument, place the tripod over the station. The legs of the tripod should be spread so that they make angles of about 60° with the horizontal. The shoes of the legs should be pushed into the ground by applying a force parallel to the legs.

Centring:

Centring is done to place the vertical axis exactly over the station mark. Approximate centring is done by means of the tripod legs. The exact centring is done by means of the shifting head or the centring device.

Levelling up:

Accurate levelling of the theodolite is done with the help of levelling screws or foot screws with reference to the plate levels. The levelling is done to make the vertical axis of the instrument truly vertical.

Focussing the eyepiece:

The eyepiece is focused to make the cross hairs distinct and clear. The following procedure is used.

- Point the telescope towards the sky or hold a sheet of white paper in front of the objective.
- Turn the focussing screw until the image of the object appears clear and sharp.

Angle Measurement by Repetition Method:

- Set up instrument at O with face left or right.
- Make the Vernier A at $0^{\circ}0'0''$ to Vernier B at $180^{\circ}0'0''$ with upper clamp to lower unclamp.
- Bisect the point A exactly by using lower clamp screw.
- Unclamp the upper, clamp to lower screw. Clamp rotate clockwise bisect B
- After bisecting B point make upper clamp to lower unclamp and read both the Vernier's.
- Swing the telescope towards A clockwise by leaving verniers unchanged. Using lower tangent
- Screws i.e. initial reading is final reading of first round.
- Apply the same procedure as per A for bisecting B swing the telescope towards A.
- After three rounds face is changed same three rounds are taken on changed face. Average
- Reading is calculated in each face. Mean of two faces is final reading.

Observations Table 3.1:

Inst. Stn.	Obs. Stn.	Face Left					Face Right					Avg. Angle
		Swing Right			Repe titio n	Angl e	Left			Swing		
		Reading					Reading			Repet ition	Angle	
Vernier A	Vernier B	Mea n	Vernie r A	Vernie r B	Mea n							
O	A											
	B											
	B											

Micro-Optic Theodolite.



Fig. 2.4 Micro-optic theodolite

Temporary Adjustments of Micro-Optic Theodolite:

The following five temporary adjustments are required.

Setting up:

For setting up the instrument, place the tripod over the station. The legs of the tripod should be spread so that they make angles of about 60° with the horizontal. The shoes of the legs should be pushed into the ground by applying a force parallel to the legs.

Centring:

Centring is done to place the vertical axis exactly over the station mark. Approximate centring is done by means of the tripod legs. The exact centring is done by means of the shifting head or the centring device.

Levelling up:

Accurate levelling of the theodolite is done with the help of levelling screws or foot screws with reference to the plate levels. The levelling is done to make the vertical axis of the instrument truly vertical.

Focussing the eyepiece:

The eyepiece is focused to make the cross hairs distinct and clear. The following procedure is used.

- Point the telescope towards the sky or hold a sheet of white paper in front of the objective.
- Turn the focussing screw until the image of the object appears clear and sharp.

Reiteration Method:

Reiteration method is preferred when several angles are to be taken at the particular point i. e. horizon closed.

Procedure:

- Select a station point O.
- Set the theodolite at O and do the temporary adjustments. The telescope is adjusted for left face right swing.
- Set the micro-meter to zero using upper clamp. Loosen the lower clamp, direct the telescope to the station point A and bisect A exactly by using the lower clamp and lower tangent screw.
- Note the micro-meter readings at station point.
- Loosen the upper clamp and turn the telescope clockwise until the point B is exactly bisected.
- Note the micro-meter readings at point B. Likewise points C and D
- Turn the telescope at station point A until the point A is exactly bisected.
- The difference of two readings will be included angle between two lines. Such as $\angle AOB$, $\angle BOC$, $\angle COD$, $\angle DOA$.
- Repeat the process to take the readings at face right condition.
- Calculate the angle of $\angle AOB$, $\angle BOC$, $\angle COD$, $\angle DOA$.
- Calculate the mean angle. Then add all the angles. It should be 360°
- If not then find out amount of correction for each individual angle.
- Compute the correct angles of $\angle AOB$, $\angle BOC$, $\angle COD$, $\angle DOA$.

Observation Table 4.1:

Inst. Stn.	Obs. Stn.	Face Light Swing Right		Face Right Swing Left		Correction	Corrected Angle
		Reading	Angle	Reading	Angle		
O	A						
	B						
	C						
	D						
	A						

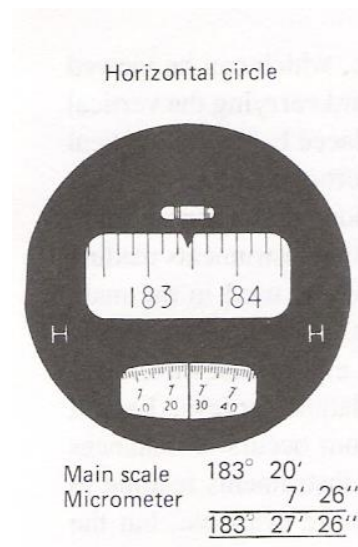


Fig. 2.5 Reading system of Micro-optic Theodolite

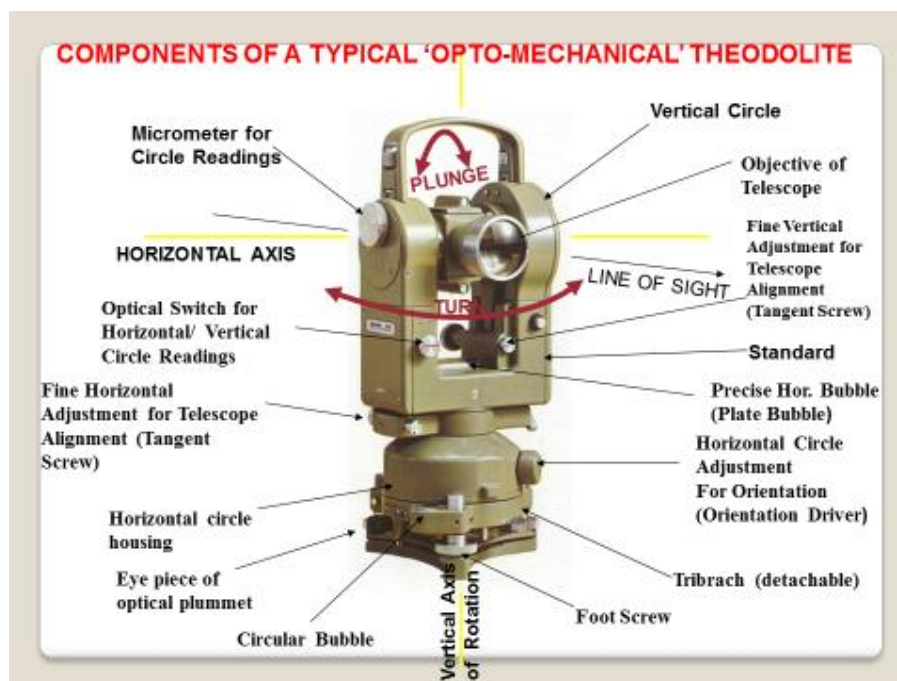


Fig. 2.6 Components of Micro-optic Theodolite

1" Theodolite:

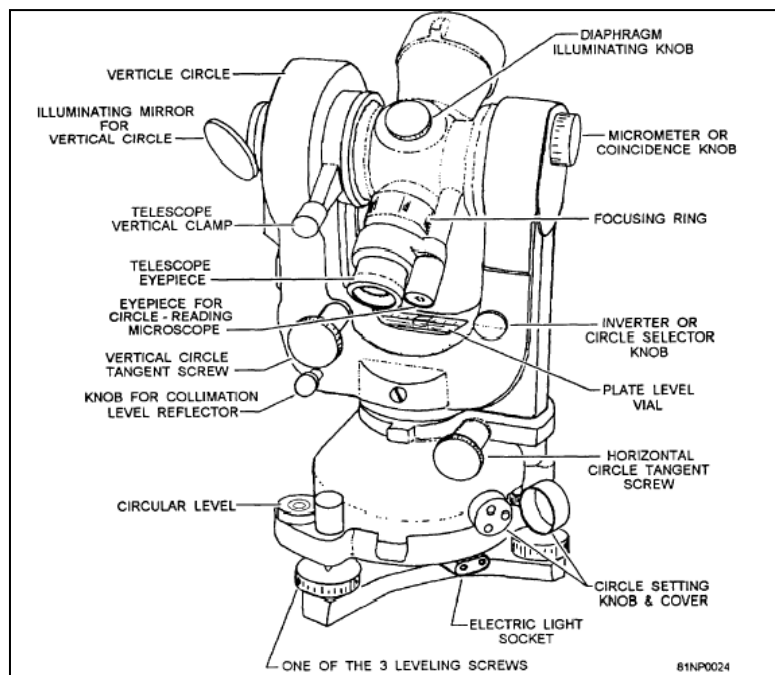


Fig. 2.7 1" Theodolite

1" Theodolite:

One second Theodolite is a precision instrument for measuring angles in the horizontal and vertical planes. Theodolites are used mainly for surveying applications, and have been adapted for specialized purposes in fields. A modern theodolite consists of a movable telescope mounted within two perpendicular axes—the horizontal or trunnion axis, and the vertical axis. When the telescope is pointed at a target object, the angle of each of these axes can be measured with great precision,

Use of Theodolite:

- Measuring horizontal and vertical angles.
- Locating points on a line.
- Prolonging survey lines.
- Finding difference of level.
- Setting out grades

- Ranging curves
- Tachometric Survey

Parts of Theodolite:

Telescope:

The telescope is mounted on a horizontal spindle called the horizontal axis or the trunnion axis. The aperture of the objective is generally 38mm. The magnification varies from 18 to 28 diameters. The length of the telescope tube is usually between 100 to 175 mm.

Vertical Circle:

The vertical circle is graduated in degrees, with graduations at 20'. The graduations in each quadrant are numbered from 0 to 90 degree in opposite directions from the two zeroes placed at the horizontal diameter of the circle.

Vertical Clamp Screw:

The vertical circle and hence the telescope can be clamped at any desired vertical angle by means of the vertical clamp screw.

Altitude Bubble:

A sensitive level tube, called altitude bubble, is attached to the top of the T-frame. The sensitivity of the altitude bubble is usually 20'' per 2 mm.

Plate Level:

A level tube, called plate level, is mounted on the upper plate. The sensitivity of the plate level is usually 35'' per 2mm.

Tripod:

Theodolite is mounted on a strong tripod when being used in the field.

Fundamental Lines: There are 5 fundamental or axis of a theodolite.

- Vertical axis.
- Horizontal axis.
- Line of collimation.
- Altitude level axis.
- Plate level axis.

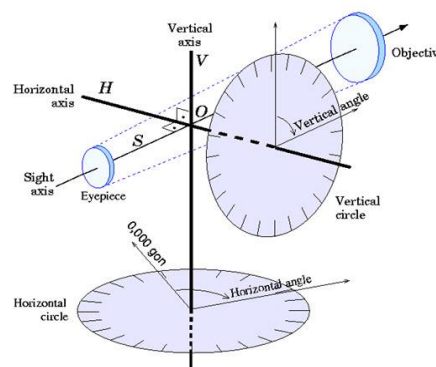


Fig. 2.8 Fundamental lines of theodolite

When the theodolite is in proper adjustment, the following four conditions between fundamental lines are satisfied.

- The axis of the plate level is perpendicular to the vertical axis.

- The horizontal axis is perpendicular to the vertical axis.
- The line of collimation is at right angles to the horizontal axis.
- The axis of the altitude level is parallel to the line of collimation when it is horizontal, and the vertical reads zero.

The horizontal axis is perpendicular to the vertical axis. It is along the trunnion axis and the line of collimation is visualised as a line along the axis of the telescope. It always remains in a plane perpendicular to the horizontal axis.

The altitude level axis and the plate level axis are perpendicular to the vertical axis.

Functions of clamp and tangent screws:

Clamp Screws:

- Horizontal clamp screw is to rotate the instrument on the horizontal plane when it is unclamped. When it is tightened it fix the instrument to the particular point.
- Vertical clamp screw is to rotate the instrument on the vertical plane when it is unclamped. When it is tightened it fix the instrument to the particular point.

Tangent Screws:

For small movement of plates, corresponding tangent screws are used. Before using any tangent screw, the corresponding clamp screw must be tightened first; it will not work.

Temporary Adjustments of a Theodolite:

The following five temporary adjustments are required.

Setting up:

For setting up the instrument, place the tripod over the station. The legs of the tripod should be spread so that they make angles of about 60° with the horizontal. The shoes of the legs should be pushed into the ground by applying a force parallel to the legs.

Centring:

Centring is done to place the vertical axis exactly over the station mark. Approximate centring is done by means of the tripod legs. The exact centring is done by means of the shifting head or the centring device.

Levelling up:

Accurate levelling of the theodolite is done with the help of levelling screws or foot screws with reference to the plate levels. The levelling is done to make the vertical axis of the instrument truly vertical.

Focussing the eyepiece:

The eyepiece is focused to make the cross hairs distinct and clear. The following procedure is used.

- Point the telescope towards the sky or hold a sheet of white paper in front of the objective.
- Turn the focussing screw until the image of the object appears clear and sharp.

Study of 1" Theodolite:

Cantering:

Cantering means setting the theodolite exactly over an instrument- station so that its vertical axis lies immediately above the station- mark.

Transiting:

Transiting is also known as plunging or reversing. It is the process of turning the telescope about its horizontal axis through 180° in the vertical plane thus bringing it upside down and making it point, exactly in opposite direction.

Swinging the telescope:

It means turning the telescope about its vertical axis in the horizontal plane. A swing is called right or left according as the telescope is rotated clockwise or counter clockwise.

Face Left:

If the vertical circle of the instrument is on the left side of the observer while taking a reading, the position is called the face left and the observation taken on the horizontal or vertical circle in this position is known as the face left observation.

Face Right:

If the vertical circle of the instrument is on the right side of the observer while taking a reading, the position is called the face right and the observation taken on the horizontal or vertical circle in this position is known as the face right observation.

Changing Face:

It is the operation of bringing the vertical circle to the right of the observer, if originally it is to the left, and vice – versa. It is done in two steps; firstly revolve the telescope through 180° in a vertical plane and then rotate it through 180° in the horizontal plane i.e. first transit the telescope and then swing it through 180°.

Line of Collimation:

It is also known as the line of sight .It is an imaginary line joining the intersection of the cross-hairs of the diaphragm to the optical centre of the object- glass and its continuation.

Axis of the telescope:

It is also known an imaginary line joining the optical centre of the object- glass to the centre of eye piece.

Axis of the Level Tube:

It is also called the bubble line. It is a straight line tangential to the longitudinal curve of the level tube at the centre of the tube. It is horizontal when the bubble is in the centre.

Vertical Axis:

It is the axis about which the telescope can be rotated in the horizontal plane.

Horizontal Axis:

It is the axis about which the telescope can be rotated in the vertical plane. It is also called the trunnion axis.

Measurement of horizontal angles by reiteration method:

This method is another precise and comparatively less tedious method of measuring the horizontal angles. It is generally preferred when several angles are to be measured at a particular station. This method consists in measuring several angles successively and finally closing the horizon at the starting point. The final reading of the vernier A should be same as its initial reading.

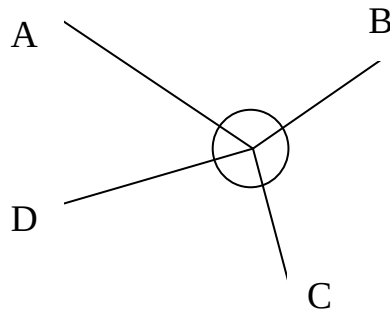


Fig. 2.9 Horizontal angle measurement by reiteration method

Procedure:

Suppose it is required to measure the angles AOB, BOC and COD.

- Set up the instrument over station point O and level it accurately.
- Direct the telescope towards point A which is known as referring object. Bisect it accurately and check the reading of Vernier and note down the initial reading. Loosen the lower clamp and turn the telescope clockwise to sight point B exactly. Read the verniers again and the mean reading will give the value of angle AOB.
- Similarly bisect C & D successively, read both verniers at each bisection, find the value of the angle BOC and COD.
- Finally close the horizon by sighting towards the referring object (point A).
- The Vernier A should now read the same initial reading. If not note down the error. This error occurs due to slip etc.
- If the error is small, it is equally distributed among the several angles. If large the readings should be discarded and a new set of readings be taken.

Observations Table:

Inst. Stn.	Obs. Stn.	Face Light Swing Right		Face Right Swing Left		Correction	Corrected Angle
		Reading	Angle	Reading	Angle		
O	A						
	B						
	C						
	D						
	A						

EXPERIMENT: 3

Name of the Experiment:

Closed Traversing using theodolite and distribution of closing errors

Traversing:

A traverse is a series of connected lines whose lengths and directions are to be measured and the process of surveying to find such measurements is known as traversing. In general, tapes are used to measure length and theodolite is used to measure the interior and exterior angles to find out the direction of traverse lines.

Types of Traverse:

A Traverse may be of two types

1. Open Traverse
2. Closed Traverse

1. Open Traverse:

A traverse is said to be open traverse when the traverse starts at one point and terminates at another point as shown in the figure. Open traverse is also called as unclosed traverse. It is suitable for surveying of roads, coastal lines, etc.



Fig 3.1: Open Traverse

2. Closed Traverse:

A traverse is said to be closed traverse when the traverse formed a closed circuit as shown in the figure. In this case, both starting and terminating points of the traverse coincide with each other. It is suitable for the survey of boundaries of ponds, sports grounds, forests, etc.

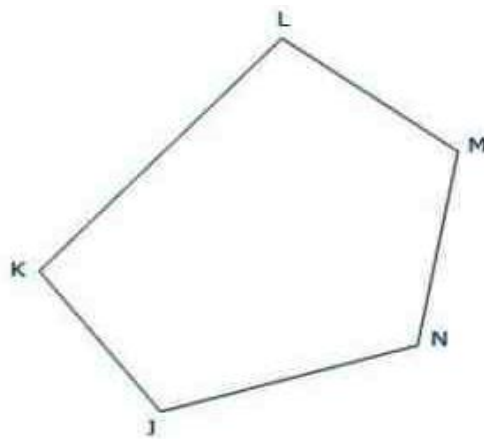


Fig 3.2: Closed Traverse

Methods of Traversing:

The traversing is performed by four different methods and these methods are classified according to the survey instrument used. The methods are as follows.

1. Chain Traversing
2. Compass Traversing
3. Theodolite Traversing
4. Plane Table Traversing

Theodolite Traversing:

In the case of theodolite traversing, the linear measurements are done by using tape and angular measurements are done by theodolite. Using theodolite, the all interior and exterior angles are measured and then magnetic bearing of the first traverse line is measured with the help of Prismatic Compass or Gyro Theodolite or Gyro Mat and from that magnetic bearing of other sides are calculated. This method is very accurate compared to other methods.

Selection of Traverse Stations:

The basic principle of a survey, i.e., working from whole to the part should be adopted. A minimum number of stations should be selected. Length of traverse lines is as long as possible reduce time and centering effect of the stations. It is better to select stations on a level and firm ground. After the selection of stations, they are marked with pegs.

Linear and Angular Measurements:

Distances between stations are measured by tape or tacheometric method or electric distance measurements (EDM) instruments like geodimeter or distameter or by Total Station Instruments. Angular measurements are done by prismatic compass or theodolite.

Plotting of Tranverse Survey:

There are two methods of plotting a traverse survey. The methods are as follows

Distance and Angle Method In this method:

Distances between stations are laid off to scale and included angles or bearings are plotted by protector or tangent of angle or chord of the angle.

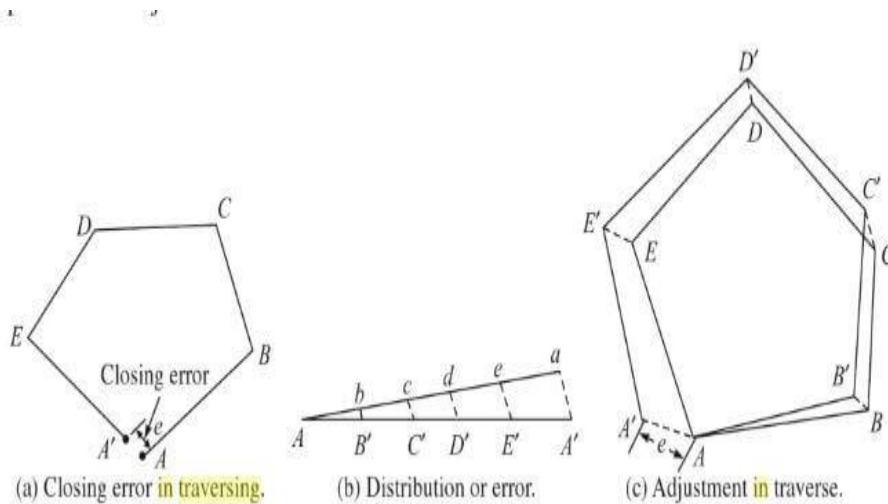
Co-ordinate Method:

In this method, survey stations are plotted by calculating their co-ordinates. This method is more practical and accurate. The biggest advantage of this method of plotting is that closing error can be eliminated by balancing.

CLOSING ERROR AND ADJUSTMENTS:

When a close traverse is plotted according to field measurements, the ends of the traverse usually do not coincide owing to the error in the measurement of angles and distances.

Errors in angles mostly occur due to local attraction. This error is called the closing error. As per below Figure (a), AA' is closing error, i.e. $AA' = e$



Adjustment

Method of closing error by graphic method:

In as per above Figure (a), the polygon, $ABCDEA'$ represents an unbalanced traverse having closing error equal to e , $AA' = e$ as the point A and last point A' are not coinciding.

The closing error AA' (e) is distributed linearly to all sides in proportion to their length by graphical construction shown in as per above Figure (b) in which AB' , $B'C$, CD' , DE' , $E'A'$ are the lengths of the sides of traverse AB , BC , CD , DE , and EA either to the same scale as per above Figure Figure (a) or in reduced scale.

The ordinate $A'a$ is equal to the closing error AA' of Figure (a). Join Aa and drop perpendiculars (or parallel lines of $A'a$) $E'e$, $D'd$, Cc , $B'b$ on line AA' as shown in as per above Figure Figure (b).

Thus similar triangles are formed and corresponding error $E'e$, $D'd$, Cc , $B'b$ are obtained. as per above Figure (c). EE' , DD' , CC and BB' are drawn a parallel of closing error AA' . Then cut $EE' = E'e$, $DD' = D'd$, $CC = Cc$ and $BB' = B'b$. Join AE , ED , DC , CB , and BA . The polygon $ABCDE$ so obtained in as per above Figure Figure (c) represents the adjusted traverse.

Bowditch's Method. :

The basis of this method is on the assumption that the errors in linear measurements are proportional to f and that the errors in angular measurements are inversely proportional to $\sqrt{l}$ where l is the length of a line. Bowditch's rule.

Also termed as the compass rule, is mostly used to balance a traverse where linear and angular measurements are of equal precision.

The total error in latitude, and in the departure is distributed in proportion to the lengths of the sides.

- The Bowditch Rule is :
Correction to latitude (or departure) of any side = Total error in latitude (or departure) x
(Length of that side / Perimeter of traverse)

Thus, if C_e = correction to the latitude of any side

C_d = correction to the departure of any side

E_L = total error in latitude

D = total error in departure

E_l = length of the perimeter

l = length of any side

$$C_l = \frac{E_L \cdot l}{\sum l} \text{ and}$$

$$C_d = \frac{D \cdot l}{\sum l}$$

Transit Method:

The transit rule may be employed where angular measurements are more precise than the linear measurements.

According to this rule, the total error in latitudes and in departures is distributed in proportion to the latitudes and departures of the sides.

It is claimed that the angles are less affected by corrections applied by the transit method than by those by Bowditch's method.

The transit rule is :

Correction to latitude (or departure) of any side

$$= \text{Total error in latitude (or departure)} \times \frac{\text{Latitude (or departure) of that line}}{\text{Arithmetic sum of latitudes (or departures)}}$$

Thus, if

L = latitude of any line

D = departure of any line

L_T = arithmetic sum of latitudes

D_T = arithmetic sum of departure

We have,

$$C_L = \Sigma L \cdot \frac{L}{L_T} \quad \text{and} \quad C_D = \Sigma D \cdot \frac{D}{D_T}$$

Advantages and Disadvantages of Methods of Plotting Traverses:

The various methods of plotting a traverse have their own advantages and disadvantages.

Plotting a traverse by the method of included angles or directions with a protractor is the least accurate but the fastest method.

The most accurate and the best method to layout a single angle precisely is by tangents.

If a traverse fails to close and if the error is due to drafting, the error may be adjusted simply by rotating each line, except the first, slightly and progressively.

In the chord method of plotting a traverse, the erection of a perpendicular (in the tangent method) is eliminated, but the determination of the chord length is more laborious than finding perpendicular offsets.

The main disadvantage of the chord method is that an error in any of the traverse line is passed through along to the succeeding line.

Plotting by independent coordinates is supposed to be a better method since the independent coordinates of the stations do not affect the succeeding coordinates and thus the error, if any, is localized to a particular point and therefore, correction usually involves the replotting of the erroneous point only.

In plotting with consecutive coordinates, small errors of plotting are carried forward, and in a closed traverse, the closing error due to inexact plotting can be located since the coordinates used have already been balanced.

This may be considered to be an advantage, as the cause of discrepancy can be searched for and eliminated.

Since errors are not accumulated when plotting with independent coordinates, they are preferred for plotting an open traverse

EXPERIMENT: 4

Name of the Experiment:

Introduction to various Levelling instruments and carryout profile leveling using auto level

Tilting level:

The telescope of the Tilting level is not rigidly fixed to the vertical spindle. The telescope can be tilted on a pivot about a horizontal axis in the vertical plane upwards or downwards through a small angle by means of a tilting screw. The tilting screws acts against a spring. A bull's eye level or pond level is fixed to the upper plate of the leveling head for approximate leveling by foot screws. Exact leveling is done using the tilting screw. The tilting screw is turned till the bubble of the level tube attached to the telescope is at the centre of the run. At that stage the line of sight is horizontal. The bubble is brought to the centre of run before each reading of the staff.



Fig.4.1 Tilting level

Temporary Adjustment of a Tilting Level:

Setting Up:

Set up the tripod at a convenient height and fixed the instrument on it.

Levelling Up:

Level the instrument roughly with the leveling screws till the circular bubble is in the centre.

Exact Levelling:

Bring the bubble of the level tube to the centre by using the tilting screw. The line of sight is exactly horizontal.

Focusing eyepiece:

First focus the eyepiece. Then point the telescope towards the object or staff and focus the objective and remove parallax. After focusing centre the bubble again with the tilting screw. The bubble must be relevelled using the tilting screw before each reading is taken.

Levelling Staff:

Levelling staff is a graduated rod of rectangular cross-section. It is a giant size ruler used for the measurement of vertical distance or elevation. The leveling staff is kept vertical at the point whose elevation is either known or is to be determined and the reading of the staff is taken from the leveling instrument. The leveling staff has the different if height. The bottom of the staff represents zero reading. Graduations on the staff give the distance from the bottom of the brass cap. Each meter is divided into 200 divisions. The thickness of each division is 5 mm. The divisions are painted alternatively black and white.

Types of Levelling Staff:

Based on the method of taking reading, the leveling staffs are divided into two types.

- Self-Reading Staff
- Target Staff

Based on the method of surveying, the leveling staff are divided into three types.

- Solid Staff
- Folding Staff
- Telescopic Staff



Fig.4.2 Different types of levelling staff

Fly Levelling:

This type of differential levelling is used for the determination of approximately elevations of different points. As it is a very approximate form of levelling, the distance as long as possible are taking. The fly levelling is done where rapidity, but low precision is required. Fly levelling is normally used for the reconnaissance of the area or for approximately checking of the levels.

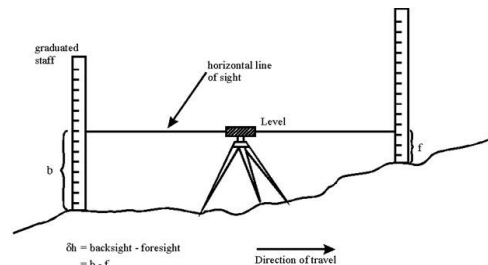


Fig 4.3 Fly levelling

Auto Level:

The line of sight of Auto Level is automatically made perpendicular to the vertical axis by an optical system called compensator. This perpendicularity is maintained even when the telescope axis is inclined to the horizontal, within centre limits of the inclination.



Fig. 4.4 Auto Level

Profile Levelling:

Profile levelling is the process of levelling along a fixed line to determine the elevation of the ground surface along that line. This is a type of differential levelling done for the purpose of determining the elevations of the ground surface along a fixed line. The fixed line is generally the centre line of a proposed route, such as a road, canal, railway line, sewer line. The profile levelling is also called the longitudinal levelling. The level obtained in profile levelling are used for plotting the longitudinal section which is required for various purposes Such as fixing the gradients, determining the earthwork quantities.

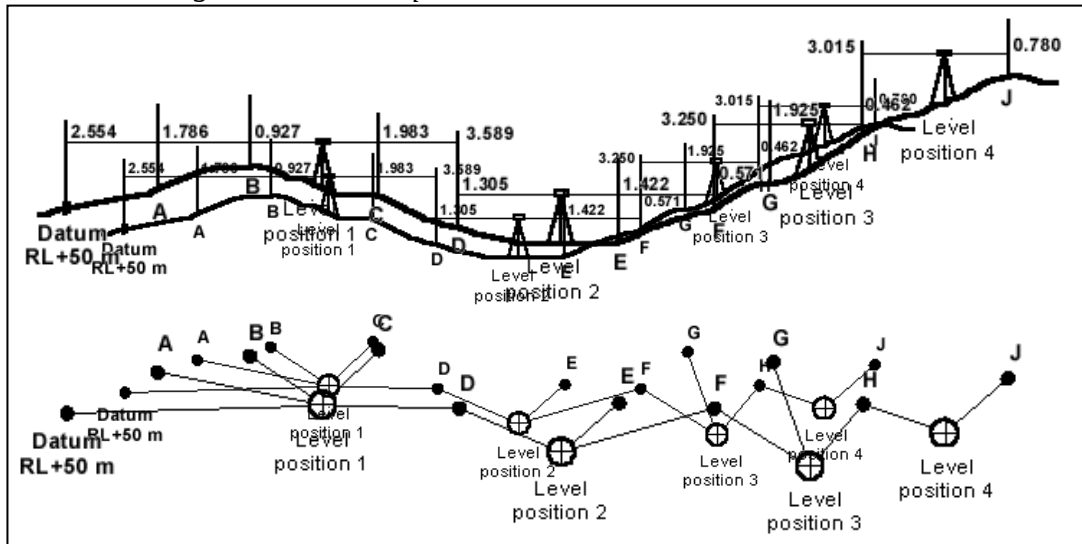


Fig. 4.5 Profile levelling

Field Procedure:

Profile leveling is nothing but differential leveling done along a line. Profile leveling should be commenced from a bench mark and should end at a bench mark, as far as possible. Let AB be the fixed line along which profile leveling is to be done. Let BM-1 and BM-2 be the two bench marks. Intermediate points a, b, c, d, e, f, g, h etc are first marked on the line AB at some suitable interval, usually 20 m. The positions of these intermediate points are located by chaining along the line AB and noting the distances from the starting station A.

The instrument is set up on a firm ground at some suitable place so as to command a large number of points on the line. A back sight is taken on BM-1 to determine the height of instrument at that instrument point. The staff is then kept successively at station A, a, b, c and d. The intermediate sites are taken to determine the level of these points. It becomes necessary to shift the instrument if the intermediate sight exceeds 100 m or so, or when further points became invisible owing to large undulations in the ground. The staff man selects a suitable turning point (TP-1) on a firm ground or on a permanent object and a foresight is taken. The turning point need not be on the fixed line AB. The instrument is then shifted and set up at point so that it commands forward points. A back Sight is taken on the turning point and the height of instrument at that point is determined. The intermediate sights are taken at point's e, f, g, h and finally B. Finally, a fore sight is taken at BM-2 to check the level.

**Standard Level Book:
Observation Table.:**

Staff Stn.	Distance	B.S.	I.S.	F.S.	H.I.	R.L.	Remarks
B.M-1		1.835			212.234	210.505	B.M-1
A	0		2.130			210.190	R.L.=210.505
a	20		2.210			210.130	
b	40		2.140			210.200	
c	60		1.980			210.360	
d	80		1.680			210.660	
T.P.		0.435		1.155	211.620	211.185	T.P.
e	100		1.580			210.040	
f	120		1.830			209.790	
g	140		2.550			209.070	
h	160		2.940			208.680	
B	180		2.830			208.790	
B.M-2				2.430		209.190	B.M-2
Sum		2.270		3.585			RL=209.190

Arithmetic Check:

$$\sum BS - \sum FS = \text{Last RL} - \text{First RL} = 2.270 - 3.585 = 209. - 210.505 = - 1.315 = - 1.315 \text{ (Ok).}$$

Plotting the Profile:

After the levels have been determined in the field, the profile is plotted. The vertical scale is generally exaggerated to show clearly the undulation in the ground. The vertical scale has taken 10 times the horizontal scale. Thus, if the horizontal scale is 1:1000, the vertical would be 1:100.

Select a suitable datum for plotting. Draw a horizontal line represent the datum line. Mark chainages of the various points along the datum line. Draw perpendicular line at each plotted chainage, and set off the respective levels at these perpendicular lines.

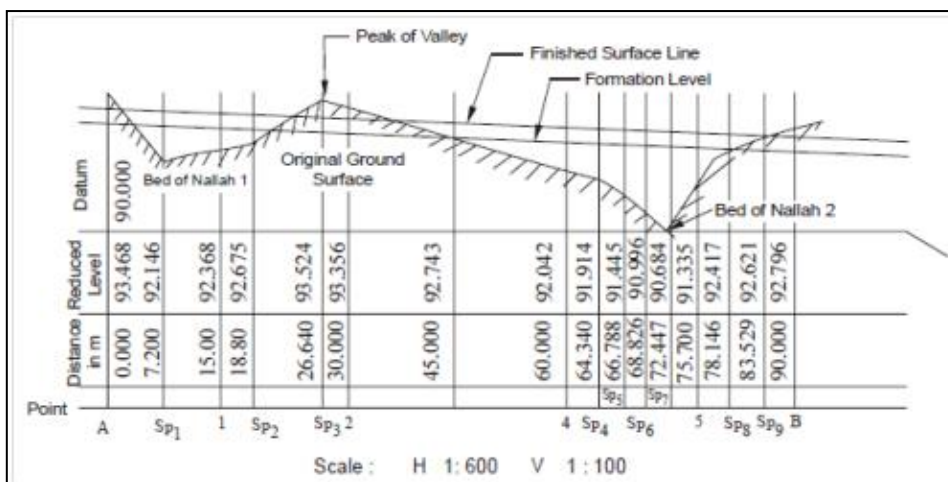
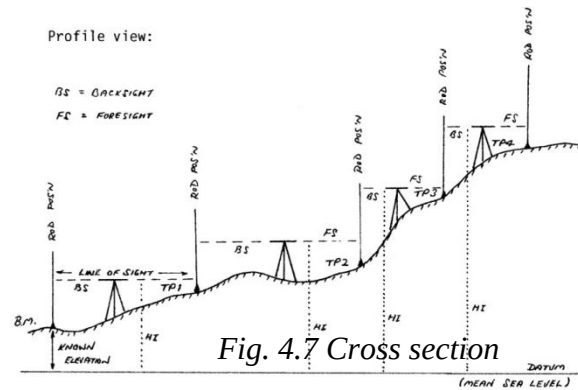


Fig.4.6 profile plotting

Cross Section:

Cross sections are run at right angle to the fixed line along which the profile leveling is done. These sections are taken on either side of the fixed line to obtain the lateral outline of the ground surface. The cross-section are plotted to depict the topography at right angles to the fixed line.



EXPERIMENT: 5

Name of the Experiment:

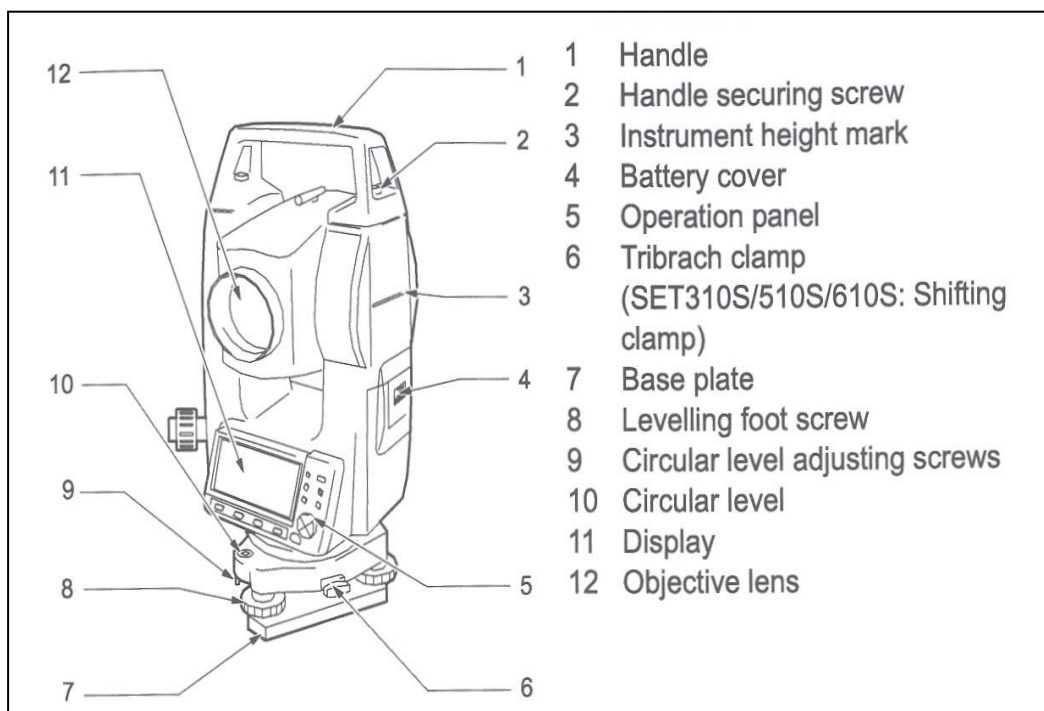
Study and sketch of Total Station and determination of coordinates using Total Station

Total Station:

It is a light weight, compact and fully integrated electronic instrument combining the capability of an EDM and an angular measuring instrument such as wild theodolite. It can perform the following functions.

- *Distance measurement*
- *Angular measurement*
- *Data processing*
- *Digital display of point details*
- *Storing data in an electronic field book*

Parts of Total Station:



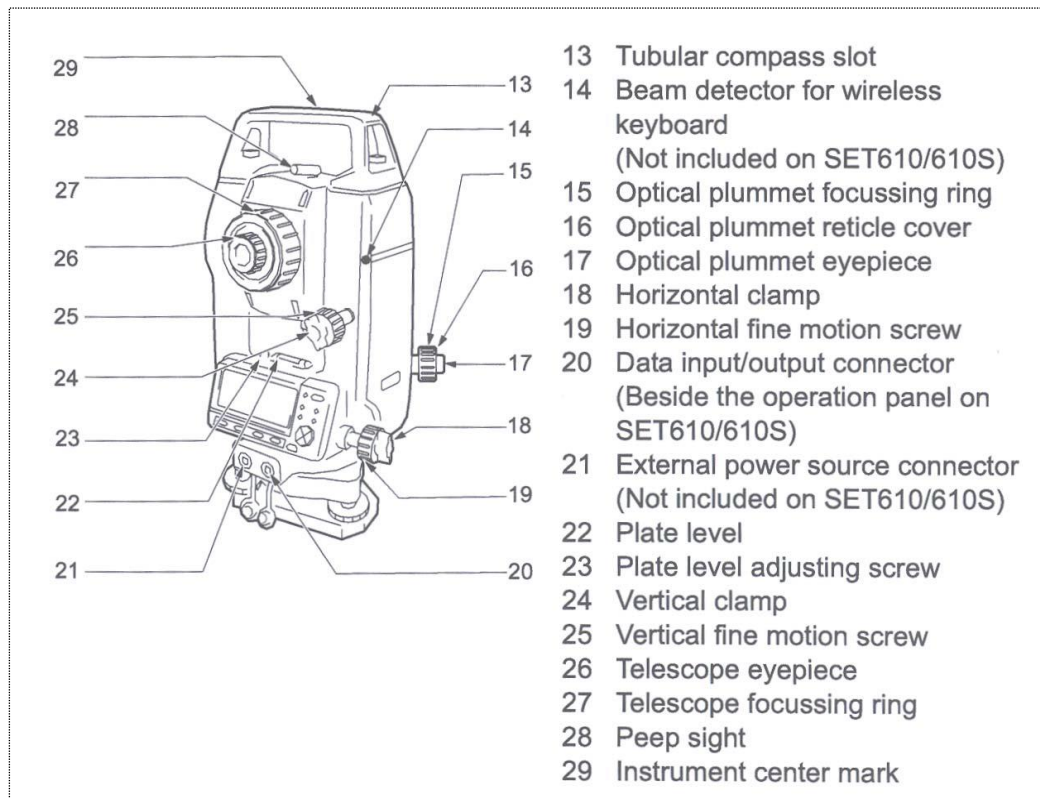


Fig. 5.1: Parts of total station

Use of Total Station:

The instrument is mounted on a tripod and is levelled by operating levelling screws. Within a small range instrument is capable of adjusting itself to the level position. Then vertical and horizontal reference directions are indexed using onboard keys. It is possible to set required units for distance, temperature and pressure (FPS or SI). Surveyor can select measurement mode like fine, coarse, single or repeated. When target is sighted, horizontal and vertical angles as well as sloping distances are measured and by pressing appropriate keys they are recorded along with point number. Heights of instrument and targets can be keyed in after measuring them with tapes. Then processor computes various information about the point and displays on screen. This information is also stored in the electronic note book. At the end of the day or whenever electronic note book is full, the information stored is downloaded to computers. The point data downloaded to the computer can be used for further processing. There is software like auto civil and auto plotter clubbed with Auto cad which can be used for plotting contours at any specified interval and for plotting cross-section along any specified line.

Distance measurement:

When a distance is measured with a total station, an electromagnetic wave or pulse is used for the measurement – this is propagated through the atmosphere from the instrument to reflector or target and back during the measurement. Distances are measured using two methods.

- The phase shift method
- The pulsed laser method.

Phase shift method:

This technique uses continuous electromagnetic waves for distance measurement. Although these are complex in nature, electromagnetic waves can be represented in their simplest form as periodic waves. The wave completes a cycle when moving between identical points on the wave and the number of times in one second the wave completes the cycle is called the frequency of the wave. The speed of the wave is then used to estimate the distance.

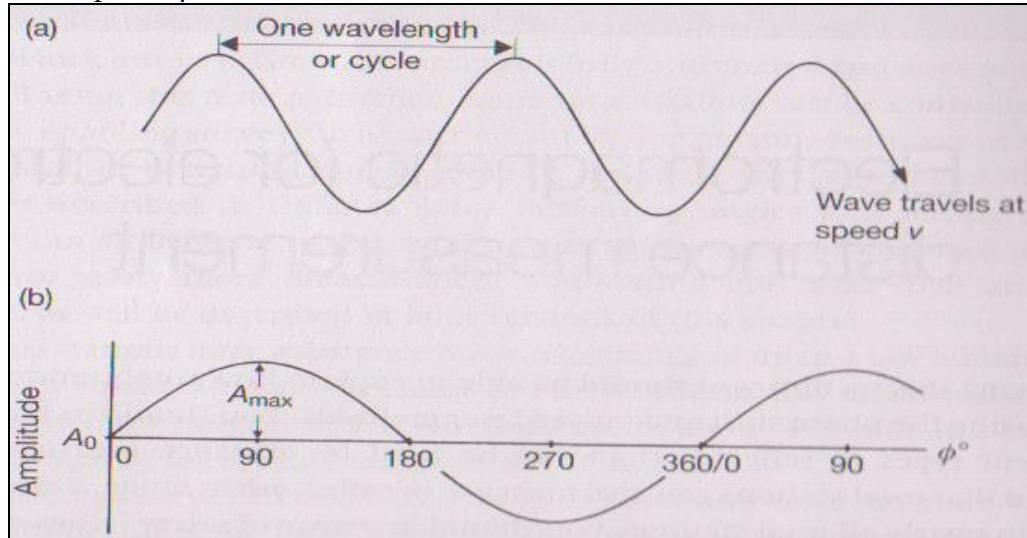


Fig. 5.2: Phase shifting

Pulsed laser distance measurement:

In many total stations, distances are obtained by measuring the time taken for a pulse of laser radiation to travel from the instrument to a prism (or target) and back. As in the phase shift method, the pulses are derived from an infrared or visible laser diode and they are transmitted through the telescope towards the remote end of the distance being measured, where they are reflected and returned to the instrument.

Since the velocity v of the pulses can be accurately determined, the distance D can be obtained using $2D = vt$, where t is the time taken for a single pulse to travel from instrument – target – instrument. This is also known as the timed-pulse or time-of-flight measurement technique.

Slope and horizontal distances:

Both the phase shift and pulsed laser methods will measure a slope distance L from the total station along the line of sight to a reflector or target. For most surveys the horizontal distance D is required as well as the vertical component V of the slope distance.

Horizontal distance $D = L \cos \theta = L \sin z$

Vertical distance $V = L \sin \theta = L \cos z$

Where θ is the vertical angle and z is the zenith angle. As far as the user is concerned, these calculations are seldom done because the total station will either display D and V automatically or will display L first and then D and V after pressing buttons.

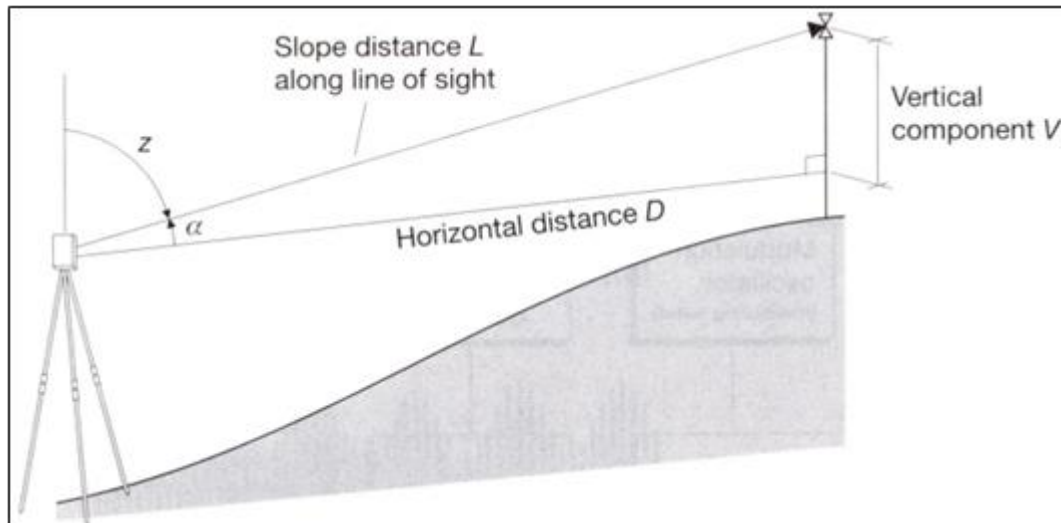


Fig. 58.3: Slope and horizontal distance measurement

Components Used in Total Station Surveying:

- Total Station Instrument
- Tripod Stand
- Electronic Notebook
- Prism P
- Prism pole
- Computer interface
- Batteries

Angle measurement with total station:

The angle measurement system of the total station is similar to that of an electronic theodolite, the dynamic angle-measuring system used in the total station makes use of a large number of graduation for each reading of the angle on Horizontal or Vertical circle of the instrument. This helps to eliminate as far as possible the effects of graduation errors. And the other parameters like horizontal distance, co-ordinates, levels etc. are also calculated automatically and displayed digitally. The display may be alphanumeric and graphic as well. A number of trigonometrical functions and setting out parameters can also be performed. Initial setting parameters like bearing of a line, co-ordinates and height of the station occupied, the height of instrument and that of the reflector height needs to be input in the instrument memory before starting the actual measurement. It can also be used in trekking mode for continuous measurement on a moving reflector for setting out points of interest.

Advantages of Using Total Stations:

The following are some of the major advantages of using total station over the conventional surveying instruments:

- Field work is carried out very fast.
- Accuracy of measurement is high.
- Manual errors involved in reading and recording are eliminated.

- Calculation of coordinates is very fast and accurate. Even corrections for temperature and pressure are automatically made.
- Computers can be employed for map making and plotting contour and cross-sections. Contour intervals and scales can be changed in no time.
- Relatively quick collection of information
- Multiple surveys can be performed at one set-up location.
- Easy to perform distance and horizontal measurements with simultaneous calculation of project coordinates (Northings, Eastings, and Elevations).
- Layout of construction site quickly and efficiently.
- Digital design data from CAD programs can be uploaded to data collector.
- Daily survey information can also be quickly downloaded into CAD which eliminates data manipulation time required using conventional survey techniques.

Disadvantages of Total Station Surveying:

- Vertical elevation accuracy not as accurate as using conventional survey level and rod technique.
- Horizontal coordinates are calculated on a rectangular grid system. However, the real world should be based on a spheroid and rectangular coordinates must be transformed to geographic coordinates if projects are large scale.
- Examples: highways, large buildings, etc.
- As with any computer-based application “Garbage in equals Garbage out”. However, in the case of inaccurate construction surveys “Garbage in equals lawsuits and contractors claims for extras.”

Aim: -Study of Total Station (TOPCON)

Instruments: - (1) Total Station (TOPCON)
Model No. GTS 211 D
Make – TOPCON
Made in Japan

Least Count: 1" (Angular), 1 mm (Linear)

Accuracy: $\pm (3\text{mm} + 2 \text{ ppm})$ Linear, 3" Angular

Range: 1200 m with single prism, 1800 m with three prisms

Storage capacity: 2000 stations

Digital display on both the sides up to 9 digits

Theory: It is a combination of linear measuring instrument as EDM, Angular measuring as Electronic Theodolite, programmed calculator for calculating coordinates (EN & Z), Data storage device in its internal memory & data communication devices



Fig. 5.4: TOPCON Total Station

Working Steps: -

- (i) Centering & leveling of the instrument at known point
- (ii) Switch on the power switch of the instrument (Topcon)
- (iii) Initiate the instrument by rotating the telescope
- (iv) Collimate the telescope toward the prism at known station.
- (v) Input Instrument station data:
 - (a) Press co-ordinate mode key
 - (b) Press F_4 to bring page 2
 - (c) Press F_3 as OCC to input co-ordinate of instrument station.
 As E:
 N:
 RL (Z):
 - (d) Press F_4 as page 2
 Press F_2 as INS.HT to input Instrument station height.
 - (e) Press F_4 as page 2
 Press F_1 as Reflector Ht and Input Ref Ht.
- (vi) Press Ang to make the instrument normal mode working.
- (vii) Collimate Back-sight station
- (viii) Input the bearing of Instrument to back sight by pressing F_3 as HSET and enter required bearing by (F_1 as Input & select numerals to Input required bearing.)

To Get Output:-

- (A) Press  (Distance meas. Key) to get

HR - Bearing
HD - (Horizontal Ditt.)
VD - Vertical Ditt.
V - Vertical Angle
HR - Bearing
SD - Slope Distance
- (B) Press again  to get



(C) Press  (Coordinate Meas. Key) to get

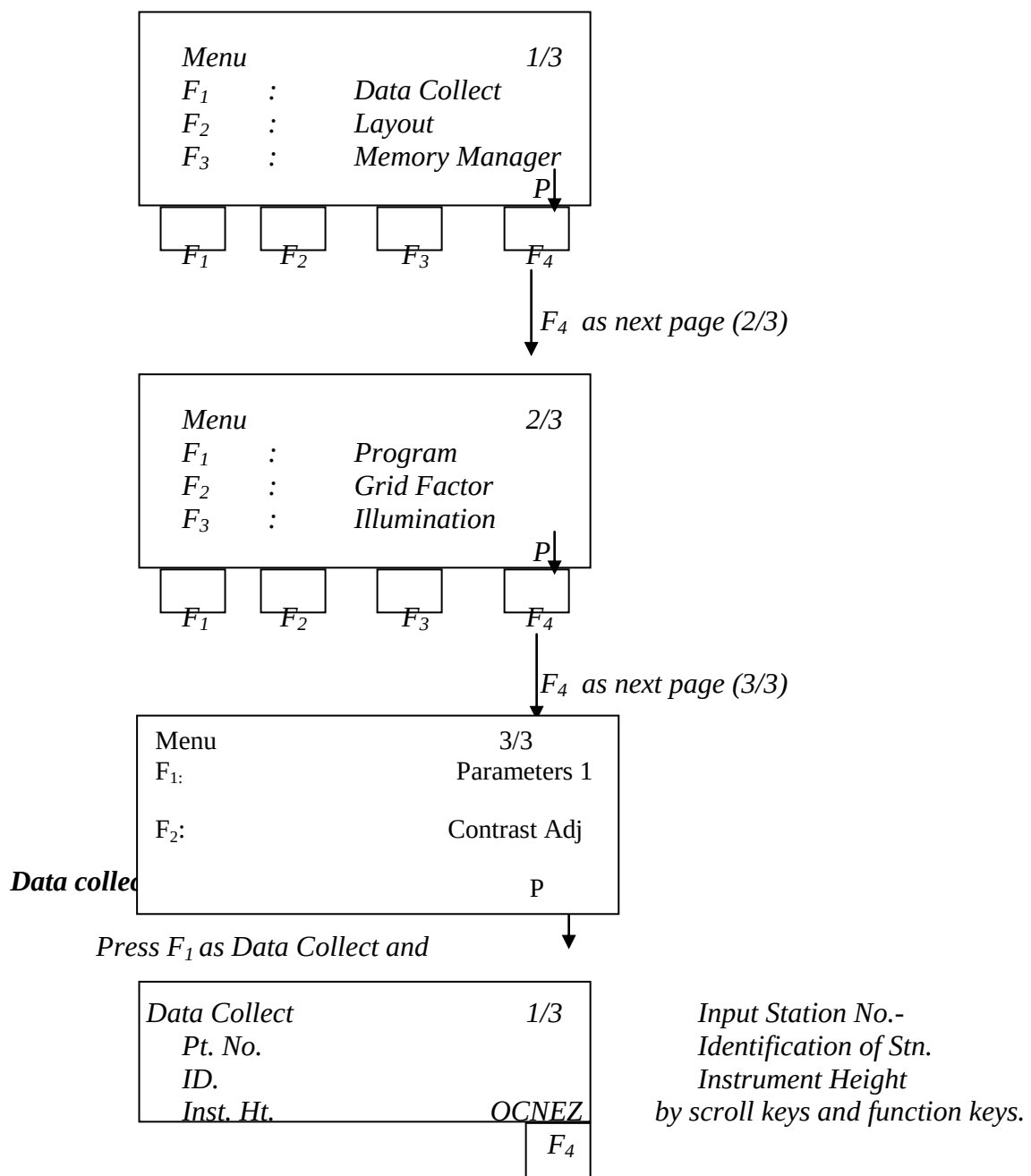
E - Easting
N - Northing
Z - Reduced Level

(D) Press Ang for normal mode.

Collimate next station and change Reflector Height (If required)
and Repeat (A), (B), (C) & D to get output.

Working with Topcon in Internal Memory: -

Press Menu Key to get difference functions of Topcon as:-



$$\boxed{ESC} \longrightarrow \boxed{ESC}$$

Press F_4 *as BS*

Press $\boxed{F_3}$ as Measure and F_2 as HD to get
 HR:
 HD:
 VD:

If required Rec ? Yes to store the data

If required Rec ? Yes to store the data.

Repeat the process for next station.

Working in Layout Parameter

In this parameter the instrument will help to locate the station on ground if Coordinate of the stations are given.

Steps are as follow:-

Centring of the Instrument at known station.

Press Menu

Press F_2 as layout

Press F_1 as Instrument Station

and input the value of Inst. Station No. &

E
N
Z

 of Inst. station

Collimate back sight station

Press F_2 as back sight

Input back sight station No. and

E
N
Z

 of back sight station

Press F_3 as layout

Press F_2 as list and explore station No. to be located using scroll key.

Press F_1 as view to see & check the co-ordinates of required station.

Press Enter and Input Reflector height if required, we will get calculated bearing & distance of the station to be located with respect to Instrument station.

Press Enter and Press F_1 as

Angle

 to get angular deviation as dHR of the line of telescope. Make dHR as $0^{\circ} 00' 00''$ and instruct the prism man to move the prism and keep it in the line of telescope.

Press F_2 as

Dist

, the instrument will measure the present distance and gives HD and dHD as difference in distance. If dHD is (-) ve then instruct the prism man to move away to the instrument depending upon the magnitude of (- ve) distance. If dHD is (+) ve then instruct the prism man to move towards the instrument but in same orientation.

If dHR and dHD both are 0 (zero) then fix that point as layed out.

Repeat the process as above by clicking

Next

 for other stations.

Press F_3 as

NEZ

 to input the new coordinate of the station and its name to be layed out.

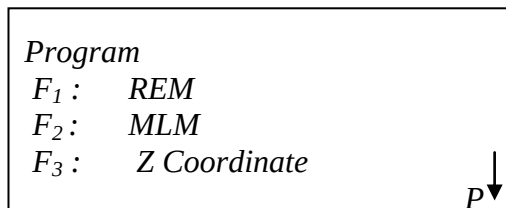
Working in Program Parameter:-

In this parameter the instrument will give Area, Remote Elevation, Missing Line and RL of the remote points from the file data or new measurements.

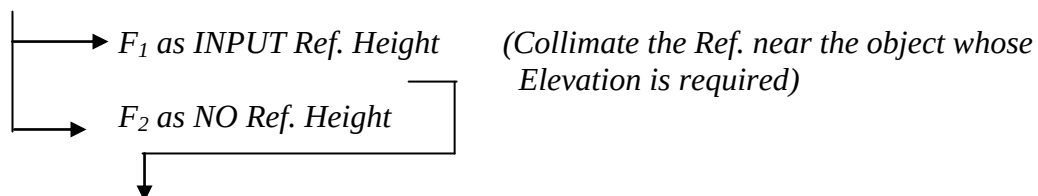
Steps are as follow:

For Remote Elevation Measurements:-

Menu → F₄ as page2 → F₁ as program



Press F₁ as REM



Case I Enter prism Height and Press F₁ as Measure.

Press F₄ as Set then collimate the point whose Height is required and get the height on monitor.

Case II

Press F₂ as NO Ref. Height

Press F₁ as Measure

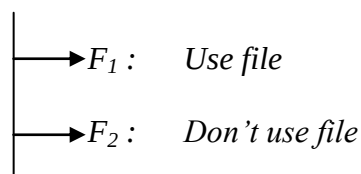
Collimate the bottom of the object and Press F₄ as set then collimate Top of the object and Press F₄ as set.

Height of the object will be on the monitor.

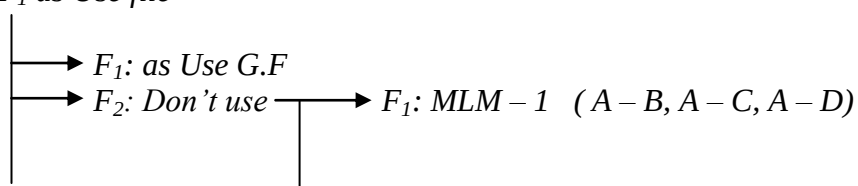
For Missing Line Measurement:

Menu → F₄ as Page 2 → F₂ as MLM

Case I



F₁ as Use file



→ $F_2: MLM-2 \quad (A - B, B - C, C - D)$

*F₁ as Meas. (After Collimation of first point as A) and Press
F₄ as set*

Collimate next station as B and Press F1 as Meas. then F₄ as
Set to get _____

d_{HD} : Horz. Dist between A & B

dVD :Level Diff between A & B.

Collimate next station as 'C' and Press F_3 as HD

Press F_4 as set to get

d_{HD} : Horz. Dist between A & C

dVD :Level Diff between A & C.

and Repeat as above for next station.

For Area Calculation:-

Case – I *Menu* $\longrightarrow$ *F₄ as page 2* $\longrightarrow$ *F₄ as page 2* $\longrightarrow$ *F₁ as Area*

F_1 : as File Data

F_2 : as Measurement

INPUT File Name or explore from list (F₂) and Enter

Press Next (F_4) by seeing the required points from list.

Area will be on Monitor after entry 3 points or more.

Case – II

Press F_2 as Measurement

Press F_1 as Meas. after collimation of first point.

Collimate the next points and

Press F_1 as Meas.

Repeat the steps up to the total points of the Polygon.

Area will be displayed on Monitor when more than 3 points measured.

Instruments:- Total Station (Sokkia)

Model No. SET 550
Make – Sokkia, Made in Japan

Least Count. 1" (Angular), 1 mm (Linear)
Accuracy: $\pm (3\text{mm} + 2 \text{ ppm})$ Linear, 3" Angular
Range: 5000 m with single prism
Storage capacity: 10000 stations
Digital display on both the sides up to 9 digits

Theory: It is a combination of linear measuring instrument as EDM, Angular measuring as Electronic Theodolite, programmed calculator for calculating coordinates (EN & Z), Data storage device in its internal memory & data communication devices

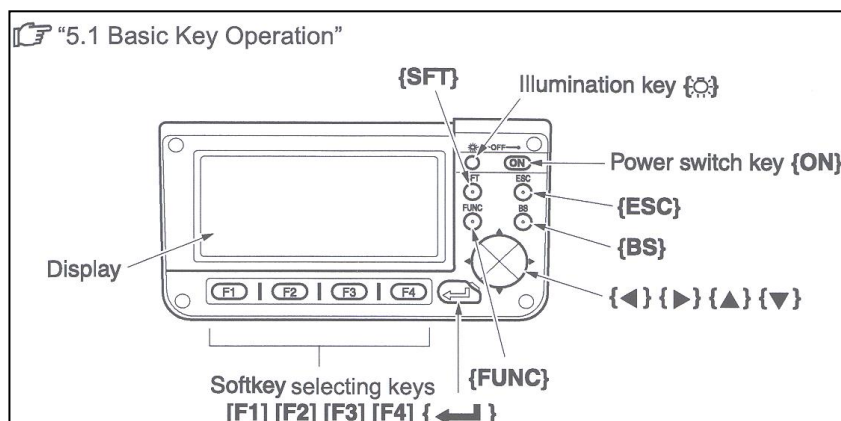


Fig. 5.5 SOKKIA Total station

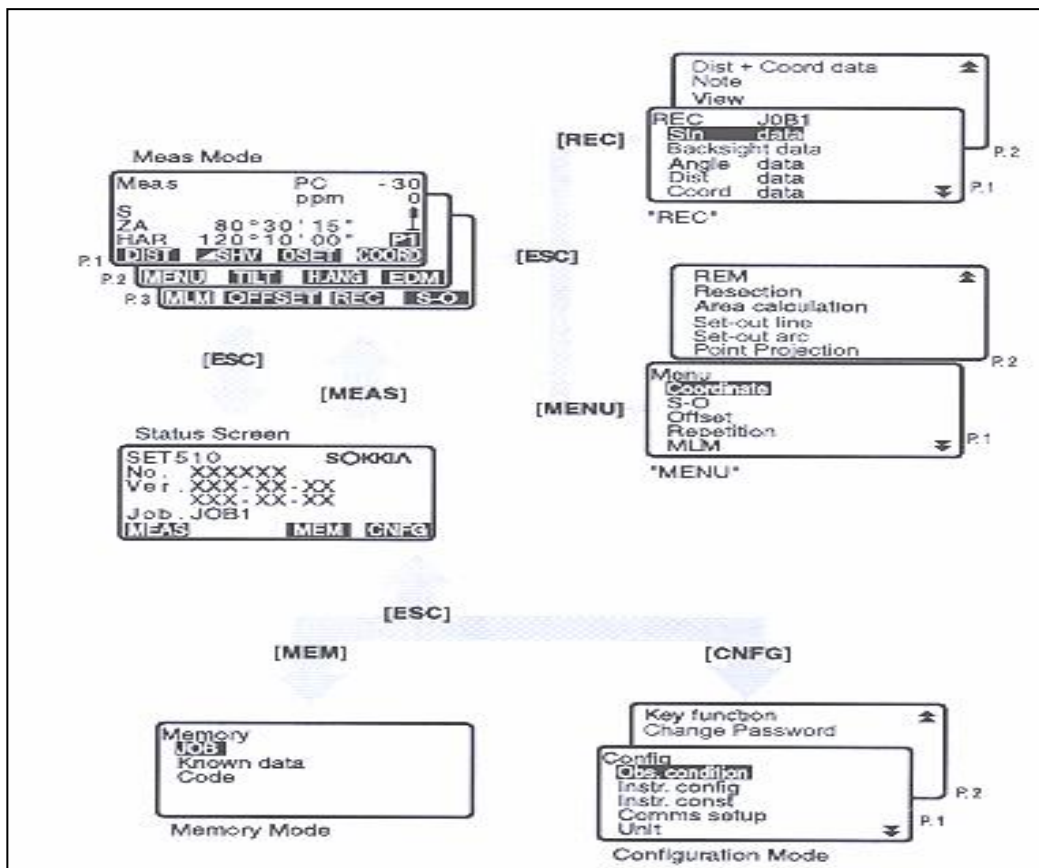
Working Steps:-

- Centering & leveling of the instrument at known point
- Switch on the power switch of the instrument (Topcon)
- Initiate the instrument by rotating the telescope
- Collimate the telescope toward the prism at known station.
- Input Instrument station data:

Basic Key Function:



Main Manu:



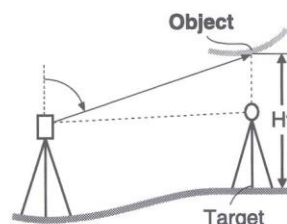
Measurement of Target Height:

- Set the Target Height from "MEAS" > "Menu" >
- "Coordinate" > "Station Orientation" > "Station Coordinate"
- Set the target height to the measured height of the mirror target. You do not have to fill out the other fields for a REM measurement
- Press "ESC" to return to the "MEAS" menu
- Select the "MEAS" > "Menu" > "REM", sight the mirror target, press [OBS] to measure "S", then [STOP]
- Sight the object above the target for height measurement
- Select [REM] and then [STOP]
- To re-shoot the mirror target use the [OBS] on the REM screen

NO:	0.000
EO:	0.000
ZO:	0.000
Inst.h	1.400m
Tgt.h	1.200m
READ REC EDIT OK	

NO:	370.000
EO:	10.000
ZO:	100.000
Inst.h	1.400m
Tgt.h	1.200m
1 2 3 4	

REM	
Ht.	6.255m
S	13.120m
ZA	89°59'50"
HAR	117°32'20"
STOP	



Measurement with Two Survey Known Points:

REM	
Ht.	6.255m
S	13.120m
ZA	89°59'50"
HAR	117°32'20"
REM OBS	

- From the "MEAS" menu
- Select [COORD] and then "Stn. Orientation".
- Set the instrument coordinates with "Stn. Coordinate"
- and then select [OK] and return to "Back-sight"
- Select "Coord" and then enter the back-sight target coordinates (NBS, EBS, ZBS) and select [OK]
- Sight in the target and inspect the "Azimuth" (it should be reasonable for your coordinate system).
- Select [YES] to calibrate.

Data down load from TOPCON to PC

Connect Data transfer cable from Topcon port to COM 1 of PC

Switch on Topcon

Initiate Topcon

Press Menu

Press F₃ as Memory Manager

Press F₄ as Page 2

Press F₄ as Page 3

Press F₁ as Data transfer

Press F₁ as send data

Press F₁ as list to select required file name to be down loaded. Use scroll key (▲▼) to select file name.

Press F₄ as Enter for selected file

Open Topcon software on PC

Click on 'COMN' as up tools

Click on download

Check for communication parameters as Boud rate, Charity/Parity on PC and Topcon.

Click on

GO

Press F₃ as yes on Topcon

Data will be down loaded to PC

Click on **OK** on text files as *.gt6

data will be saved on C drive by default.

Click on save and enter file name for the above data, as ABC

Open Micro soft Excel

Open file as ABC data will be saved in Excel.

The image shows three sequential screenshots of the Topcon instrument's LCD display. The first screen shows the 'Backsight' menu with options 'Angle' and 'Coord', where 'Coord' is highlighted. The second screen shows the 'Backsight' menu with data entry fields for 'NBS : 1.000', 'EBS : 1.000', and 'ZBS : 0.000', with 'READ', 'EDIT', and 'OK' buttons at the bottom. The third screen shows the 'Backsight' menu with 'Take BS' selected, displaying 'ZA 90° 12' 34"', 'HAR 123° 12' 34"', and 'Azimuth 45° 00' 00"', with 'REC', 'NO', and 'YES' buttons at the bottom.

Name of the Experiment: Detailed surveying and contouring with Total Station

Instrument: Leica Total Station (TS15)

Software: SURPAC 6.5

Total Station in the combination of an Electronic distance meter, electronic theodolite and a programmed calculator and may be said to be an electronic tachometer theodolite.



Fig 5.6: Leica Total Station

Specification:

Accuracy Hz, V 1''
Display resolution 0.1''
Compensator setting accuracy 0.5''
Range 5000m with Prism
Range 1000m Reflectorless

Salient Features:

- Dynamic angle Scanning System ensures highest measuring accuracy.
- With standard telescope.
- Two axis compensators automatically determine and corrects standing axis tilt.
- The data can be recorded and stored in a Card or Internal memory of the instrument, and can be downloaded to or up loaded from a computer. The data is recorded in a format compatible to Software programme and computer evaluation of the data, so that the output can be obtained in desired usable form.
- Online control by computer
- Total Station can be operated with a multi-function Keyboard, which is associated with a display screen which in some cases is giving a graphic display also.
- Built-in Co-ordinate geometry functions for ancillary calculation.
- For Special applications, compensator and corrective computations can be switched off.

Method: The Horizontal distance and horizontal & vertical angles to target are measured automatically at a push of button. Conversion into rectangular coordinate as X, Y, Z i.e. Easting, Northing and Reduced Level.

Observation:

A number of functions, settings and adjustments to set up the Total Station for specific tasks. Co-ordinates, height, and horizontal direction (angle) of instrument stations can be stored in the theodolite and are then available as station co-ordinates for the next traverse stations.

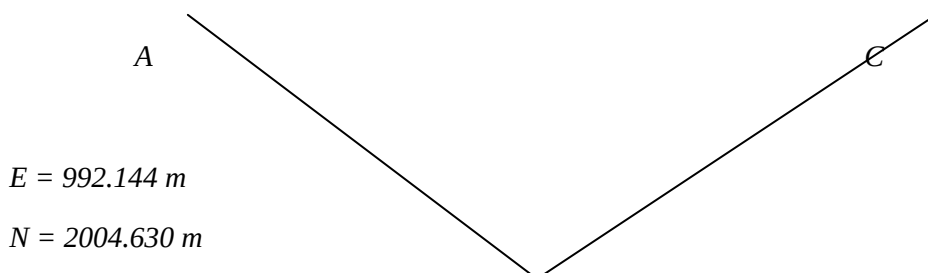
Procedure for observation by Total Station: -

(1) Data Collection for Reference Line and Occupied Station Point

Every Back-Sight Observation here required input data or Set Data in the Total Station at the occupied instrument station or traverse survey started from any survey point or any point of survey work are given below:

(1) $HR =$ Reference Line Back Bearing or known Back Bearing of the Line BA = $300^{\circ} 30' 45''$

(2) Occupied Survey Point B



$$Z = RL = 249.471 \text{ m}$$

$$B \quad E = 1000.000 \text{ m}$$

$$N = 2000.000 \text{ m}$$

$$Z = RL = 250.000 \text{ m}$$

(3) Co-ordinate at the instrument station B

▪	Easting = E	=	1000.000 m
▪	Northing = N	=	2000.000 m
▪	Reduced level = RL	=	250.000m

(4) Instrument Height = INSHT = 1.500 m

(5) Height of Reflector or Prism = R. HT = 1.450 m

(2) PREPARATION FOR MEASUREMENT:

By putting prism on standard pole and moving on the survey stations in that vicinity E, N, Z determined in total station accordingly saved in the job created earlier.

3. Data transfer:

Data Transfer from Total Station to Computer may be done by USB port available in TS. Pen drive can be used to transfer the data

4. Contouring using Surpac

4.1 GRAPHICS-BASED METHOD

- Open string file in graphic pane
- Surface > Create DTM from a layer

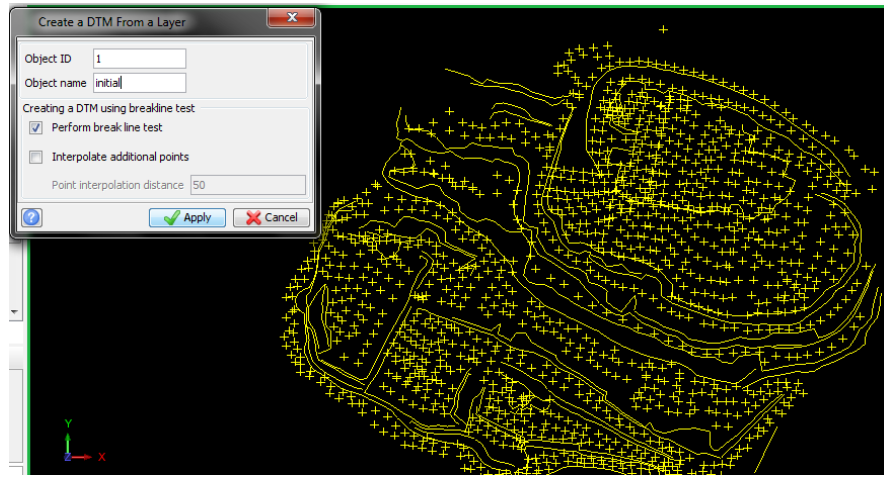


Fig. 5.7: x, y, z points

(If there are crossovers then zoom in, move points or break the segment to remove it, then repeat the process from starting)

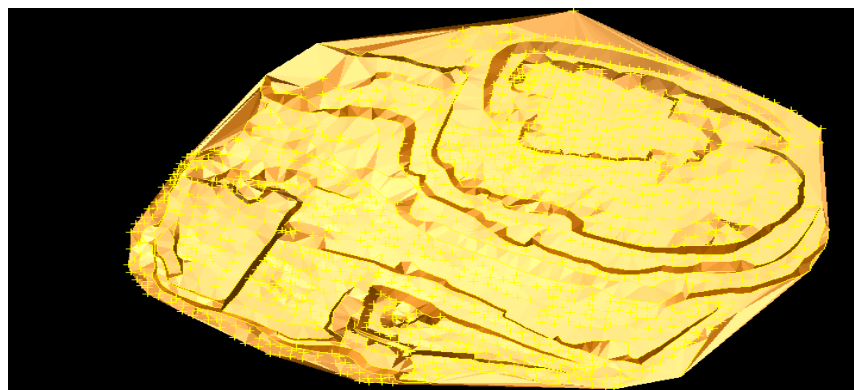


Fig. 5.8: DTM

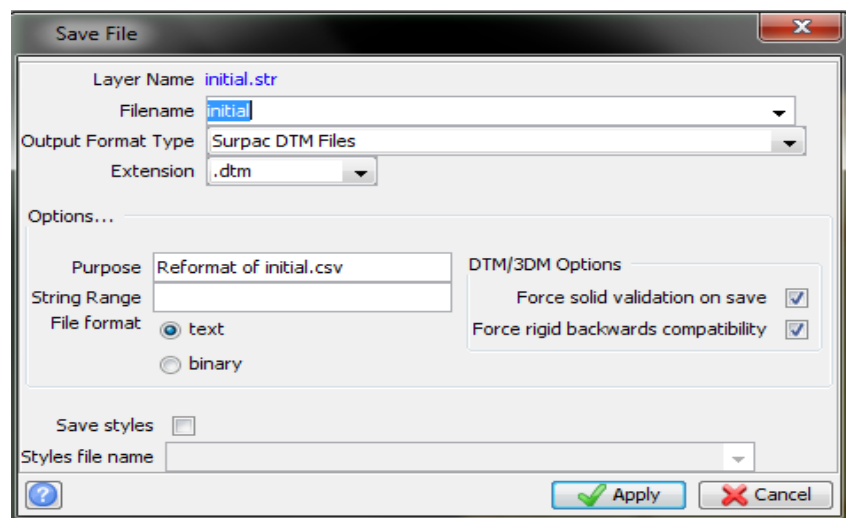


Fig. 5.9

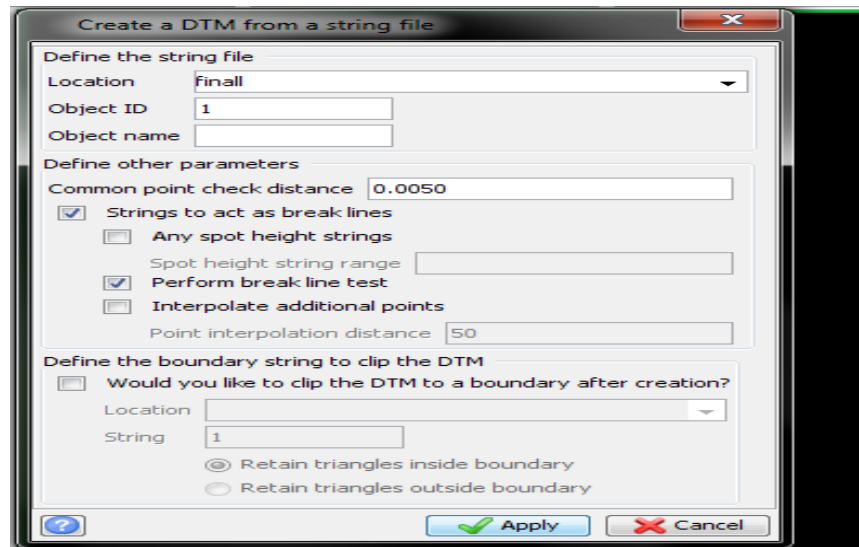


Fig. 5.10

(DTM is created & the log file opens in your default text editor.

Open saved .dtm file.



Fig. 5.11

APPLICATIONS OF DTM CONTOURING

*Surfaces > Contouring > Contour DTM in layer.
Select interval Or Select range*

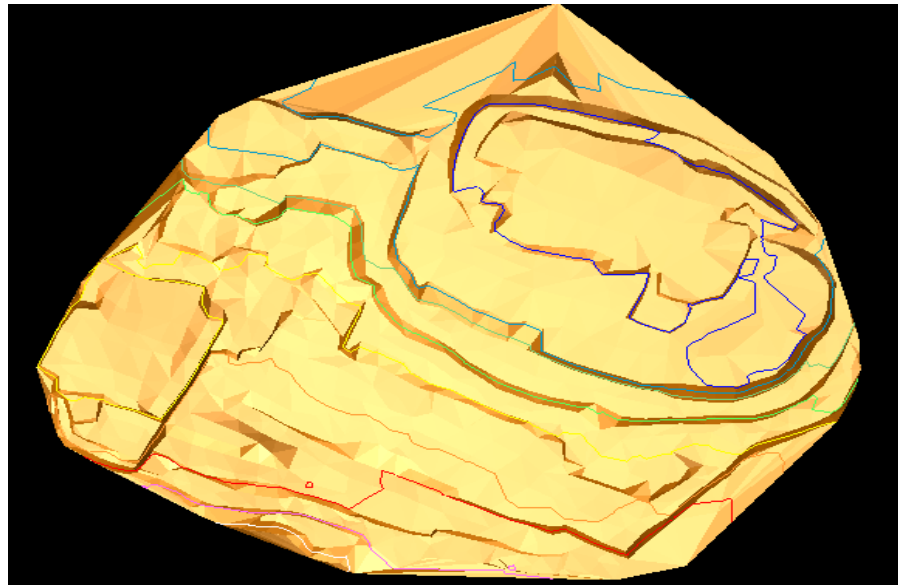


Fig. 5.12

Active the layer > Save the layer by clicking right.

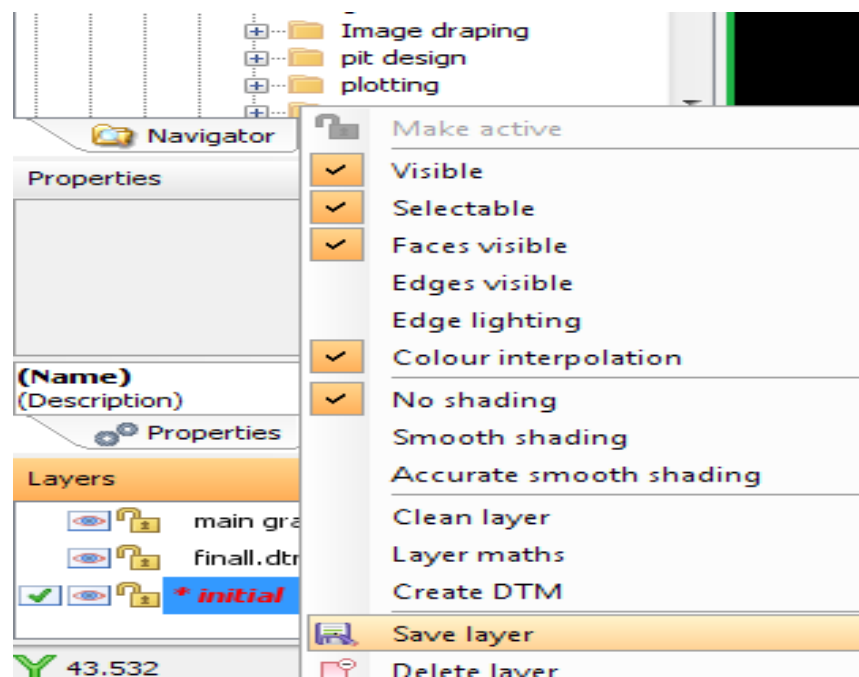


Fig. 5.13

Open the saved str file.(To know z values)
Display > points > attributes

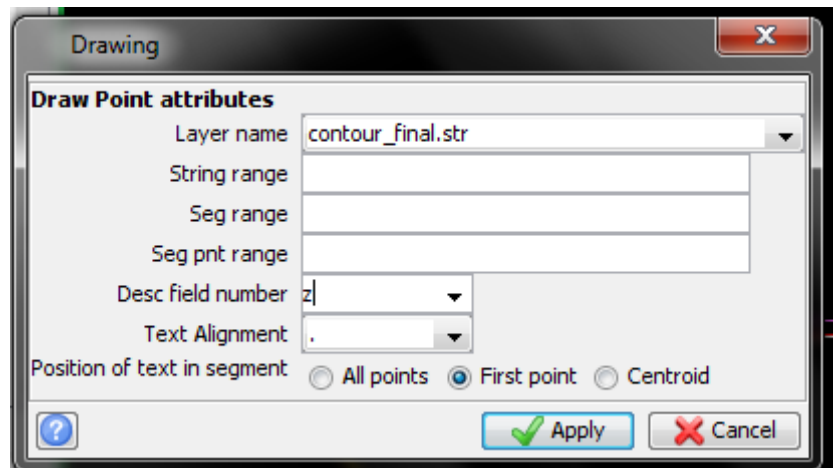


Fig. 5.14

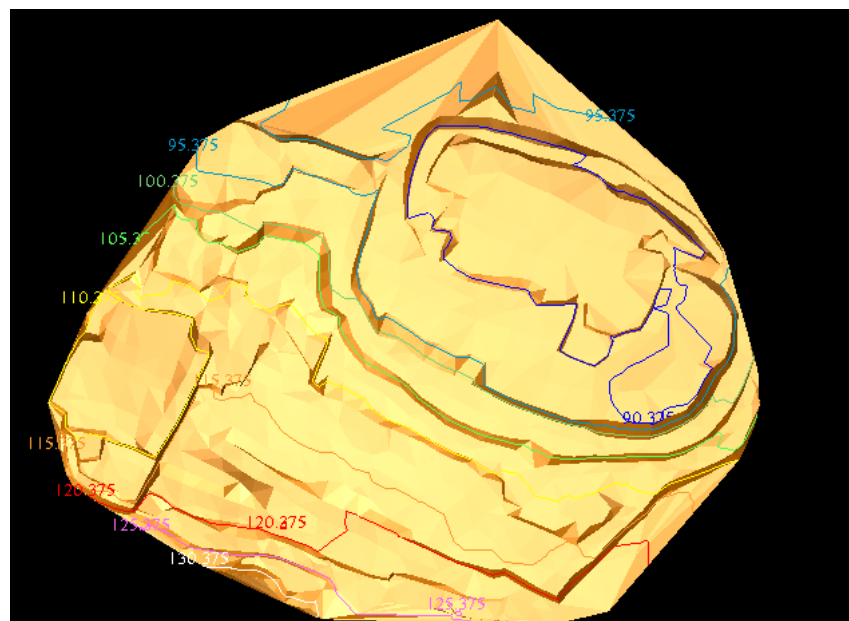


Fig. 5.15

Precaution:

1. Once the centering and leveling is complete the instrument should not be disturbed.
2. The instrument should be handled with care.
3. The telescope must be activated before feeding reading.
4. Before switching off the systems, all files opened must be closed.

5. *The instrument should not be disturbed after collimation of the instrument with the prism is completed.*

EXPERIMENT : 6

Name of the Experiment:

Preparation of digital mine plans using Total station data

Downloading the data from TOPCON GTS 211D to PC

Connect Data transfer cable from Topcon port to COM 1 of PC

Switch on Topcon

Initiate Topcon

Press Menu

Press F₃ as Memory Manager

Press F₄ as Page 2

Press F₄ as Page 3

Press F₁ as Data transfer

Press F₁ as send data

Press F₁ as list to select required file name to be down loaded. Use scroll key () to select file name.

Press F₄ as Enter for selected file

Open Topcon software on PC

Click on 'COMN' as up tools

Click on download

Check for communication parameters as Boud rate, Charity/Parity on PC and Topcon.

Click on

GO

Press F₃ as yes on Topcon

Data will be down loaded to PC

*Click on **OK** on text files as *. gt6*

data will be saved on C drive by default.

Click on save and enter file name for the above data, as ABC

Open Micro soft Excel

Open file as ABC data will be saved in Excel.

Downloading the data from SOKKIA SET X to Pen Drive:

- *Switch on the instrument and fix the pen-drive or card.*
- *Press program bottom.*
- *Press function.*
- *Press **COM***
- *Press **COM***
- *Check the file should be Yes.*

- *Function control then enter.*
- *Data format in Delimited text.*
- *Click send.*
- *Select Job*
- *Choose Job and make it Yes.*
- *Press enter, after complete the download then click Ok.*
- *Double click in removal disk / storage card.*
- *Input the Job name, then .txt*
- *Now click **OK***
- *Then the down lording is complete*

Downloading the data *from SOKKIA SET X to Computer:*

- *Open **SOKKIA** link*
- *New Project or exiting project*
- *Go to data.*
- *Total station*
- *Point **N,E,Z** code (coma)*
- *Open data.*
- *Choose down load file and open it*
- *Click on Drawing then OK*
- *File*
- *DXF file*
- *Save the DXF file then OK*
- *Choose the destination folder and input the file name then OK.*
- *Now close the SOKKIA link.*

Preparation of grid and plotting the field data

Field Data Plotting with LIS CAD Software

Plotting of .sdr File:

- *Go to file and click on New*
- *Browse the folder where data is stored*
- *Give the file name and click on ok*
- *Go to Task and Click on Field Transfer*
- *Go to Resolve and Click on Create Field File*
- *Select .sdr file and click on ok*
- *Select the Device (Sokkia SDR33) and click on ok*
- *Go to Resolve and click on Reduce Field File*
- *Select .fld file and click on ok*
- *Close lis CAD report window and click on yes*
- *Go to display and Select the parameter as required and click on ok*

Plotting of .csv File:

- *Go to File and click on New*
- *Browse the folder where data is stored*
- *Give the file name and click on ok*

- Go to Task and Click on Field Transfer
- Go to Resolve Click on Create Field File
- Select .csv file and click on ok
- Select the Device (Custom csv) and click on ok
- Go to resolve and click on Reduce Field File then Select .fld file and click on ok
- Close lis CAD report window click on yes
- Again Close lis CAD report window
- Go to display and Select the parameter as required then click on ok.

Plotting of .str File:

- Go to File click on New
- Browse the folder where data is stored
- Give the file name and click on ok
- Go to Task click on Data Conversion
- Go to Import click on Add/Remove
- Select Surpac and click on Add
- Close the window
- Go to import click on Surpac
- Select .str file and click on ok.

Create Plot File and Preparation of Grid:

- Display the data
- Go to task click on CAD output
- Go to Setting click on Grid
- Give value of East & North interval
- Go to Setting click on CAD System
- Lis CAD CAD for Windows
- Go to output click on fit
- Give required Scale
- Select paper then click on option and
- Tick (✓) on grid and then click on ok and ok.

Create DTM & Contour

- Plot the data
- Go to Task click on Terrain Modeling
- Go to Modeling click on Form Modeling
- Give model name and click on ok and ok
- Go to Display click on Feature
- Then click on Model and
- Tick (✓) on Triangle, Contour, Values, & Display along contour and click on ok.

EXPERIMENT : 7

Name of the Experiment: Study and sketch of electronic gyro- theodolite (Gyromat-3000) and determination of Gyro-north

Instrument:

1. Precise Level, Model No NI007, Make- Carl Zeiss, LC 0.05mm
2. Precise Staff
3. Tripod



Fig. 7.1 Zeiss Precise level No. Ni007

Theory:

Zeiss Self Aligning Auto set precise level is basically different from conventional levels. Such instrument a certain amount of tilt eliminates by the help of suspended opto mechanical unit termed tilt compensator, which is a pentaprism swings freely within the telescope path of rays. The pentaprism is secured to a pendulum of half the focal length. The Pendulum of NI007 is air damped within the swinging range of ± 10 minutes of arc. Fig. 1 is a representation of the optical components and of the arrangement of the pendulum.

The resulting target image is upright and unreserved. The pentaprism breaking down the bundle of rays, which enter the instrument horizontally it behaves as a parallel plate micrometer. Tilt is affected by turning micrometer drum the measurable amount of displacement of the line of sight is 5 mm. The cylindrical, periscope like shape of the instrument is conditioned by the optical principal adopted and by the length of the pendulum. It offers the advantage that the sighting line runs higher above the ground than with levels with horizontal telescope. The measurement will be less affected by retraction disturbances near the ground. In heavy traffic the view of the staff will be less obstructed.

Construction:

For precise leveling with the NI007 and internally suspended pentaprism behaves as a parallel plate micrometer, Inver leveling staff and fitted on the telescopic tripod.

- i) **Salient Features of NI007:** On the telescope body of NI007 are located the cover in front of the pentaprism, the focusing knob, azimuth clamp Lever and Tangent Screw, micrometer drum (with micrometer division) with clamp screw and reading magnifier, the circle reading microscope, the illuminating mirror for reading the graduated circle and the telescope eyepiece, preliminary leveling is to the circular spirit level, which is easily viewed through a prism. A parallax free sighting device serves for roughly sighting the staff.

The Micrometer provided with the instrument has a range of 5 mm, which is divided, into 100 equal parts. In precise leveling work the use is made of wedge-shaped section of the gratitude, which allows the setting of diverging hairs with almost constant accuracy for reading the staff kept over different distance. The telescope magnifications is 31.5x, the instrument is also available with a horizontal circle for setting, the instrument is a desired direction approximately with in 1' of an arc.

- ii) **Inver Leveling Staffs:** A three-meter precision leveling staff with Inver tape, graduated to 5 mm, upright numbering, and thickness of graduated lines 1.6 mm. There is a check scale on the Right Side of the Inver tape and main scale on the left side of the Inver tape. Displacement between the both scales is 60.65 cm. The mean error of graduation is claimed to be ± 0.01 mm. The least count of the micrometer reading of the instrument is 0.05 mm directly and 0.005 mm by estimation.

- iii) **Tripod:** Telescopic tripod legs in recommended for precise leveling.

Operation procedure of NI007 precise leveling instrument in precise leveling:

Setting up Instrument NI007

Setting up the Instrument NI007 on the tripod legs and the legs must be firmly into the ground. The tripod head should be as horizontal as possible and the height should be as requirement of the observer's eye level

Leveling Up

Turn the fool screws until the circular bubble lies in the center of it setting circle. The line of sight is then automatically set in horizontal by the compensator.

Focusing and Sighting

The telescope is pointed towards a uniformly light surface or a sheet of white paper, and the telescope eyepiece is turned until the vertical cross hairs appear sharp and absolutely black.

Turning the instrument by hand, the telescope is pointed roughly at the levelling staff. Turn the focusing knob, which has coarse and fine motion until the staff image appears sharp and free from parallax with

respect to the cross hairs, i.e. there should be no apparent movement between the horizontal hair and a staff graduation.

Observation of Inver Staff Reading

Now the reading on an Inver staff is, turn the micrometer knob until the wedge shaped of the reticle straddle a graduation line i.e. the graduation line is symmetrical within the wedge. 5 mm are read from the staff and part of a 5 mm are taken from the micrometer.

The staff reading taken at each set up should include tow pairs of level readings on main scale and two pairs on the check scale as they provide immediate arithmetical check on readings in the sequence as given below.

Back Staff	-	Check Scale
Back Staff	-	Main Scale
Fore Staff	-	Main Scale
Fore Staff	-	Check Scale

Displacement between the both scales is 60.65 cm.

Application:

Surface and underground bench mark levelling.

- Subsidence levelling in mining region.
- Precision leveling in Civil Engineering and in large machinery construction.

Advantage:

- (i) They are compact and lighter than old instrument. As well as precise levels are more accurate and at the same time quicker also.
- (ii) The larger magnification of telescope and the measurements by the pentaprism, which is behaved as a parallel plate micro meter, have enabled the levels to read smaller fractions and therefore has enhanced the accuracy.
- (iii) The instruments with the compensators are quicker and at the time there is ease in reading because of erect image.
- (iv) All glass to air surface of the optical system are antireflection coated and therefore there is minimum loss of light and the images are bright.

GyroMAT3000

Specification:

S.No.	Parameters	Values	
1.	Measuring mode : 2 modes minimum	Mode 1	Mode 2
2.	Measuring accuracy in mgon Standard deviation to DIN 18723, $\pm\sigma$	1 or better	10 or better
3.	Measuring time in minutes (approx.)	10 or less	2 or less
4.	Battery capacity for single measurement	25 or better	50 or better
5.	Operating temperature	-10°C to +50°C or better range of temperature	
6.	Operating range	up to 80° latitude or more	



Fig. 7.2: GyroMAT3000 along with Leica Total Station

3. Principle of Gyro

The gyro attachment consists of a gyro motor suspended vertically, like a plumb-bob on a thin metal tape. Its spin axis is therefore held horizontal through gravity. The gyro motor powered by the accompanying battery unit spins at 22000 revolutions per minute about this horizontal axis and tries to maintain, in space, its initial random spinning plane created by its moment of inertia. The gyro, together with the theodolite and tripod is earthbound and is pulled out of its original spinning plane by the earth's rotation. The gravity influenced gyro reacts to this interference and its spin axis oscillates about the plumb line until the spin axis is oriented in the meridian plane and the rotation of gyro corresponds to the rotation of the earth from west to east. As a result there is no interference, but the gyro does not stabilize immediately in north-south direction and oscillates about the meridian plane because of its mass inertia.

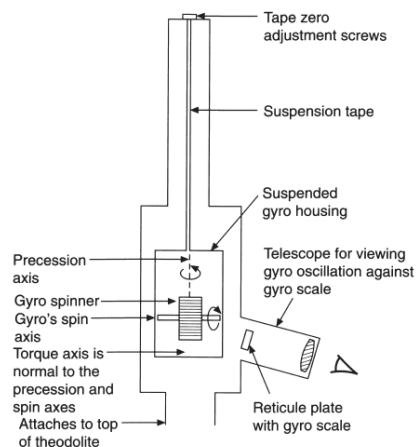
If θ is the moment of inertia of the rotor with w_K the angular or spin rate of gyro and w_E the spin rate of the earth respectively, ϕ , the latitude of the place and A , the angle of deviation of gyro's spin axis from the meridian (fig.1) then the equation for R , the directional couple forcing gyro into the meridian is given by

$$R = \theta \cdot w_K \cdot w_E \cdot \cos \phi \cdot \sin A$$

When the spin axis stabilizes in the meridian, both A and therefore the couple R are zero. In high latitudes also R decreases rapidly and becomes zero at the poles where every position of the spin axis indicates north-south direction.

4. Design

Different manufacturers have different design of the gyro attachment depending upon the type of gyro. A model called GAK 1 manufacture by M/s Wild, Heerbrugg which is much in use in surveying.



Dispositif de lecture de l'écart angulaire de l'axe de la toupie :

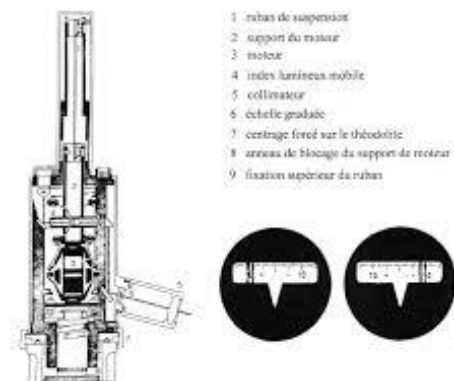


Fig.7.3 Internal components and view of Gyro

5.1 Preparations for Measuring:

The setting up, centering and leveling of the theodolite are carried out as usual. An error in the leveling up of the theodolite (in seconds of arc) will produce an error in the True North determination of about half the size. Because of this, the warning is given that the tripod must be set up carefully and firmly and that the instrument must be protected from strong sunlight. The horizontal drive screw (tangent screw) of the theodolite must be centered before measuring begins so that the same range is available in each turning direction.

Before starting observations, orient the telescope as closely as possible to North by any of the usual methods, such as the position of the sun, map, compass or a theodolite tubular compass. If a theodolite tubular compass is used the orientation must be made before the set up as the gyro contains magnetic parts.

Open the holding ring of the bridge by turning the ring to its right-hand limit. Set the gyro attachment on the bridge with the theodolite in the face left position (vertical circle to the left of the theodolite's line of sight) and with the opening for the observation tube of the gyro above the telescope eyepiece. Then, immediately turn the holding ring to the left to its limit in order to secure the gyro attachment. The holding ring must be tightened only very gently. Strong tightening of the holding ring must be avoided, as this would put all parts under strain.

Open lid of the converter and pull out the connection cable, Unscrew the protective cover of the socket on top of the gyro attachment, plug in the connection cable and fasten it by screwing up the plug's locking ring. A groove ensures that the plug and socket are connected correctly. As a precaution against strain, the weight of the cable may be supported by tying the cable to the tripod. Attach the observation tube to the tube of the housing and secure the tube by turning it to the right.

Turn the battery switch on the control panel of the converter to INTERN or, if an external battery is being used, to EXTERN. If the voltage of the battery is sufficiently high the pointer of the voltage indicator moves to the right into the white sector. The gyro scale will now be illuminated. The brightness of the illumination can be regulated by the rheostat.

5.2 *Running up and releasing the Gyro:*

It is important that the gyro is in the clamped position before it is started, therefore, first of all check that the red warning band cannot be seen. Turn the starter switch on the control panel of the converter to RUN. During the running up period, which takes 1 to 3 minutes depending on the ambient temperature, the battery is under maximum load. Check, therefore, that the pointer of the voltage indicator remains in the white sector. If it does not stay in the white sector the charge of the battery is insufficient. During running up, the gyro speed indicator will be red. When the gyro motor reaches the measuring speed this indicator changes to white.

After the gyro speed indicator turns white the gyro may be released. Look through the observation tube and turn the milled rim of the clamping device to the left until a notch, which can easily be felt, is reached. In this half-open, damping position, three damping feelers touch and rub the underside of the damping plate of the oscillating system and, through friction, damp the oscillation of the gyro. After waiting for between 5 to 10 seconds in this damping position, turn the milled rim of the clamp slowly to the left until the clamp reaches its stop and while turning the clamp, look through the observation tube and watch the gyro mark. If the gyro mark starts to vibrate strongly or if, from the beginning, the mark moves too quickly, the clamp must be turned back to the damping position (i.e. the notch). After several seconds in the damping position turn the clamp slowly to the left again until the stop is reached once more. With the clamping device in this position the gyro hangs completely free. The initial impulse given to the gyro when releasing the clamp and the angle between the gyro axis and the meridian.

5.3 *Pre-orientation:*

Normally, by using a compass, the sun or a map, it is possible to pre-orient the telescope of the theodolite towards North to within the $\pm 30^\circ$ required for a quick orientation. However, even if

none of these possibilities is available, it requires only a small amount of operating experience with the GAK₁ in order to be able to find the required pre-orientation by means of the gyro itself. When training is being given in the observing techniques of the gyro, pre-orientations should be practiced, not only starting in the quadrants adjacent to North, but also starting with the initial point of the telescope will away from North and even with it pointing to south.

At this stage it may be helpful to note, that the optical system projects the mark on the scale on the opposite side of the instrument. Thus when the gyro or rather the north end of the gyro's spin axis, swings, to the West, the gyro mark is seen moving towards right across the scale and vice-versa.

When the gyro is released with the telescope pointing more than 30° away from north, a very pronounced acceleration or slowing down of the gyro mark will be seen after only a few seconds of following up (i.e. turning the alidade by hand and keeping the mark within the v-shaped index), from which it is immediately obvious that the pre-orientation of the telescope deviates from North by a large amount. The gyro should now be clamped and the telescope swung through 30° to 45°, either in the direction of the Gyro's acceleration or away from the direction in which the gyro is slowing down. The gyro is released again and the movement of the mark observed once more. By this procedure a pre orientation to within + 20° to + 30° can be found within two minutes of time.

If the initial pointing of the telescope is to South, or close to South, the situation is not as simple as the gyro may be in a liable position. If, after about three minutes of following up, the gyro continues to accelerate and does not slow down at all, it should be clamped and the telescope turned through at least 120° from the initial telescope pointing in the direction of the gyro's acceleration. If the initial pointing is south, it is also possible that the gyro will slow down in the beginning and reach a reversal point, followed by an acceleration only. After three minutes, at the most, of continuous acceleration, the gyro should be clamped and the telescope turned through 120° from the initial pointing in the direction of the gyro's acceleration. In both cases, the new starting position will then lie in one of the quadrants adjacent to North and this position can be improved by a further attempt at pre orientation.

5.4 Quick Orientation by observing two Reversal Points:

If the telescope of the theodolite is pre – oriented to within $\pm 30^\circ$ of True North, the quick method outlined below, gives an orientation to True North with an accuracy of about $\pm 3'$ in a few minutes of time. The method is suitable therefore for orienting the telescope close to North prior to a precise orientation.

Pre-orient the telescope to within + 30° of North. Release the gyro as described earlier, slacken the horizontal clamp of the theodolite and follow up the gyro mark by turning the telescope by hand and keeping the moving mark within the V-shaped index. As a reversal point is approached the mark slows down. When the gyro mark is almost stopped, tighten the horizontal clamp of the theodolite and, by turning the horizontal tangent screw of the theodolite, keep the gyro mark exactly on the zero line of the scale (Fig. 4) until the reversal point is reached. At the reversal point itself the mark is at a standstill for a short time. Now take the horizontal circle reading U_1 corresponding to the reversal point. Slacken the horizontal clamp of the theodolite and again follow up the gyro mark, which is now moving in the opposite direction, until the second reversal point on the other side of North is reached.

Take the circle reading U_2 for the second reversal point. Clamp the gyro and compute the North direction from the mean of the two circle readings.

$$N = \frac{1}{2} (U_1 + U_2)$$

The method can be speeded up if the telescope is intentionally set to about 10° East or West of the estimated North direction. After releasing the gyro, which runs off to North, bring the gyro mark

back again by turning the clamp to the right to a little past the half-way notch of the damping position and then release the gyro again immediately. By means of this procedure the gyro will reach a reversal point within one minute of time. With most gyros, it will be noticed that obtaining a reversal point quickly, in the manner just described, is easier on one particular side of the North than on the other.

5.5 Working Procedure of Gyromat 3000:

The following steps are used for the observations of Azimuth using Gyromat 3000

a. Preparation for Measurement

- *Setup the tripod right above the base point.*
- *Level the tripod with the tripod bubble.*
- *Insert the GYROMAT into the tripod ring.*
- *Plug in the data cable between GYROMAT 3000 and Total Station /*
- *Theodolite.*
- *Pre-level the instrument with the oil bubble of the Total Station / Theodolite by turning the GYROMAT's foot screws.*
- *Switch on the Total Station / Theodolite and choose the levelling menu showing the horizontal level by the electronic inclinometers.*
- *Level the instrument as horizontal as possible by turning the three foot screws. Control this operation with the electronic inclinometers in the display of the Total Station / Theodolite in two opposite positions by turning the Total Station / Theodolite by 200gon (180deg).*
- *Centre the instrument with the mechanic plumb to the base point under the tripod or use the optional Optical Centring System.*
- *After centring control the horizontal position of the instrument again.*
- *Leave the levelling menu of the Total Station / Theodolite and choose the measurement menu.*
- *Now the GYROMAT 3000 is ready for operation.*

b. Measurement menu

- *Enter the E – factor as determined previously*
- *Start of the measurement and automated set measurement in Mode 1 measurement.*
- *Once the automated measurement is over, sighting to the target is done in both face twice in each face.*
- *The Azimuth of line of sight is displayed accordingly*
- *The measurement is repeated for 3-4 sets so as to take the mean of Azimuth of the line of sight.*

6. Table:

Gyro observations- Transit Method

Instrument No.

R. O. at

Date:

Observer

Recorder

Reading to (M) =

Reading to approx. north $N' =$

SL. No.	Time	Transit time (t)	Δt	Amplitude (-1,-1)		Remarks and calculations
				+	-	

7. Limitations

Although a gyro-theodolite functions at the equator and in both the northern and southern hemispheres, it cannot be used at either the North Pole or South Pole, where the Earth's axis is precisely perpendicular to the horizontal axis of the spinner and the meridian is undefined. Gyro-theodolites are not normally used within about 15 degrees of the pole because the east-west component of the Earth's rotation is insufficient to obtain reliable results.

EXPERIMENT : 8

Name of the Experiment:

Study and sketch of Nadir Plummet and LASER and demonstration for mine correlation survey

Instrument:

1. Nadir Plummet
2. Laser Eye Piece
3. Laser regulator and battery



Fig 8.1 Nadir Plummet

Construction:

The Nadir Plummet: -

The Nadir plummet consists of a counter mast which is to be centered above the station. The station mark can be observed with the help of a suspended prism inside the instrument. The eye piece of optical plummet can be rotated 36° around the axis of the instrument.

Laser Eye Piece

The laser eyepiece is an accessory, which could be used with theodolites, levels, plummets, and Penta prism. It interchanges with the normal telescope eyepiece converting the instruments instantly into laser theodolites, laser levels and laser plummets.

The laser light, led to the eyepiece via a fiber optic, is diverted by a Beam splitter and brought to a minute focus point at the center of the reticule, so that a pin point of light in fact a reflection is seen at the intersection of the cross hairs. The result is that the outgoing laser beam coincides exactly with the line of sight and focuses with the telescope to fall as a bright red dot on the observed target. This dot can be

distinguished in day light up to 200 m whilst in darkness ranges of 400 m are easily reached. Fitting a diaphragm to the telescope objective causes the laser beam to produce a diffraction pattern of concentric rings. This allows the center point to be picked up easily at any distance along the beam.

A built-in filter affords the observer's eye complete protection against laser light and a polarization filter cuts out glare so that the pin point of light on the cross hairs can be seen distinctly.

Theory:

Lasers are one of the biggest achievements made in the second half of the twentieth century. Lasers are quantum generators working in the optical region of the spectrum. By laser we mean, the light amplification by stimulated emission of radiation. Various substances are used for generating the lasers such as crystals, activated glasses, plastics, gases, liquids, semiconductors.

The operating wavelength of the present-day optical quantum generators extends from 0.4 to to 7 μ . The advantage of laser devices resides in a number of remarkable properties of laser radiation. In contrast to light emitted by conventional source, this radiation is coherent with speed and time. It is monochromatic, propagates in the form of a very narrow beam and is characterized by an extremely high concentration of energy, which only recently seemed fantastic. All this makes the laser beam finest instruments in the hands of scientists and engineers for the investigation of various substances. Laser beam can be used to transmit signals for communication both on earth and in space practically over any distance.

➤ Electrical and laser data

Laser source	laser diode
Wave length	635 nm
Input DL2	12 V DC
Supply voltage for intensity regulator	9V – 16V DC (rated 12 V)
Output voltage of regulator	12 V DC
Current load	80 mA – 100 mA

Procedure:

In conventional method the wire is suspended from top of the shaft to transfer the co-ordinates from surface to ground. With increase in depth this may lead to incorrect correlation. Thus, lasers are used which are very precise even at large depths. Laser beam is sent to the shaft bottom for 4 different positions of plummet at 90° apart.

Due to some disturbances of errors we may sometimes get 4 different points close to each other lines are drawn joining all the points and the center is formed. For greater accuracy works laser is projected from 8 points by turning plummet through 45°.

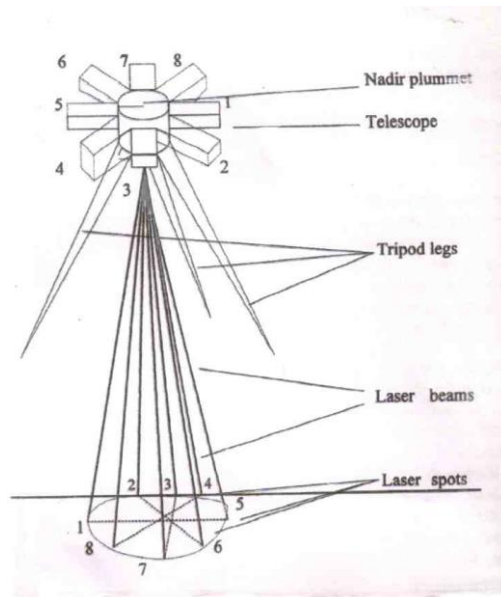


Fig.8.2 Determination of mean position of Laser spot at pit bottom

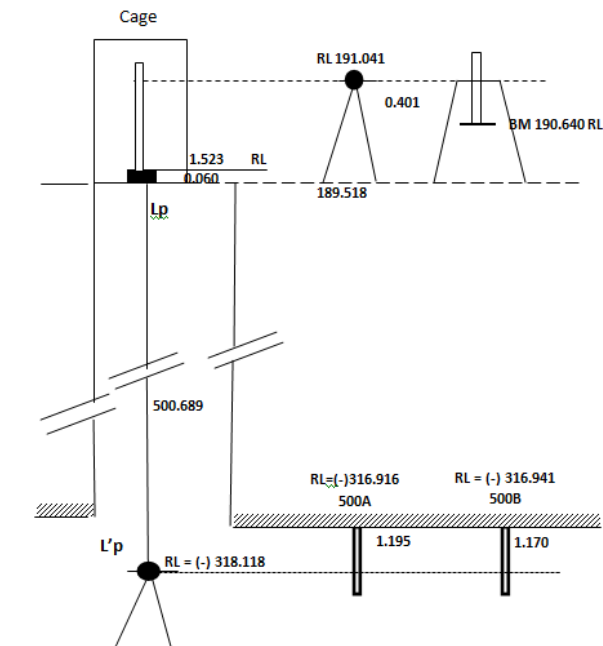


Fig 8.3 Shaft depth measurement using Total Station at pit bottom

Transfer of Direction

The bearings are transferred to the underground with the help of gyro-theodolite. The expected accuracy was $\pm 20''$. Today gyro instruments are available which can be used with the total station in place of a theodolite and have measuring accuracy of $\pm 20''$ (Sokkia – Fig. 4) and $\pm 3''$ (Gyromat 3000, Fig. 5).



Fig 8.4 A hole cut in the floor of the cage at pit top for laser beam to pass

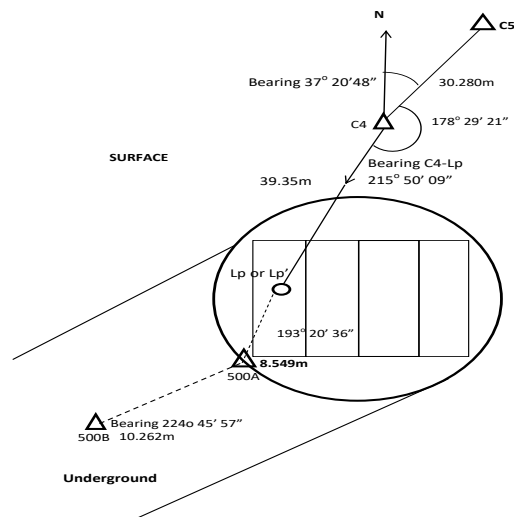


Figure 3: Correlation for transfer of coordinates

Fig 8.5 Correlation for transfer of coordinates

In the time to come when we go for greater depths for exploiting coal or minerals, this exercise will surely help to the level that a depth of 1km can very well be tackled in two stages by carrying out the exercise once from the surface level and in second stage from about 500m depth level.

APPLICATION IN MINE SURVEYING: -

1. Laser range finds: distance measurement
2. Alignment & grading
3. Plumbing (Correlation Survey)
4. Marking (Drilling & Cutting)
5. U/G Surveying a Night Surveying

Experiment: 9

Name of the Experiment: Study of GNSS system and carryout GNSS survey in relative point positioning.

Instruments:

1. GPS Base Sensor Model No. SR 510
2. GPS Rover Sensor Model No. SR 510
3. Antenna
4. Antenna Transfer Cable
5. Terminal
6. Battery
7. Tripod
8. Roving Pole
9. Base (Tribrach)
10. Antenna Pole Stud
11. Height Hook



Fig. 9.1 DGPS SR 510

Principle:

Global Positioning System gives the location of the GPS Base or GPS Rover Antenna point in term of Geodetic Co-ordinates or Cartesian Co-ordinates with the help of satellites signals orbiting the earth in different orbits.

The system works on three basic segments:

- I. Space Segment
- II. Control Segment
- III. User Segment

I. Space – Segment: Space Segment contains 32 Satellites orbiting in 6 orbits with 12 hours orbit period. All the Satellites are more than 22000 km far from earth surface. Satellites have atomic clocks which measure the time very precisely in the fraction of 10^{-9} sec; the atomic clocks control the GPS signals which have frequency of 10.23 MHz very accurately. Frequencies are modulated in two codes: -

- (a) Precision code: - For defense purpose

(b) Coarse Acquisition code: - For Civilian

II. Control – Segment: The control segment has a Master control station located near

Colorado Springs in the US and a number of tracking stations around the world. The control system has following functions: -

- (a) Monitoring the satellite transmission continuously and accurately,
- (b) Positioning the satellite for next 1 hour, based on extrapolation of their path,
- (c) Updating the navigation message periodically, and
- (d) Calibrating the satellite clocks.

III. User – Segment: The user segment comprises of the total user community who observe and record the transmissions of the GPS satellite and obtain the receiver's antenna position, velocity and time by using solution algorithms. The receiver receives the satellite signals and obtain its real time position for navigation purpose or more accurate for geodetic surveying by observing over a certain time.

Thus, simultaneous working of all three segments, a coded messages (signals) transmitted by satellites reach to the receiver with respect to time gapping.

The speed of the signal (radio waves) is constant as C i.e 3×10^8 m/s:

Distance is being measured by signal in between user receiver and satellite by following ways:-

Distance = time x speed

Or $D = TC$ where T = Time taken by signal from satellite to Receiver.

C = speed of the signal as 3×10^8 m/s

By getting the distances with respect to minimum four nos. of satellite and intersecting all distance arc's, an unique point will be determine on earth's surface that is receiver's station. Since all the bodies as satellites and earth are moving the navigated data has some error. To minimize this error observation time should be more as preferably 1 hour and the data should be processed with a processing software (Ski Pro) to minimize the ambiguities.

Observation: There are two basic classification of GPS Data observation

- (i) GPS field work which gives navigated point position of receiver station.

- (ii) GPS Data downloading and processing which gives the accurate position value in mm accuracy.

Working steps for DGPS field work: -

- (i) Set the tripod, tribrach, Antenna carrier, Antenna, Antenna cable, GPS Sensor, Terminal, battery etc. at the survey station.
(ii) Switch on the Terminal (wait up to main menu)
(iii) Click on **survey** as F_1
(iv) Press F_1 as **New Job**.

Job Name :-

Creator :-

Description:-

- (v) Press F_1 as **Cont**
(vi) Press F_1 as **Survey**

Enter Configuration set -

Job name -

Coordinate system -

- (vii) Press F_1 as **Cont**

Enter Point I D :

Antenna Height:

- (viii) Wait upto $GDOP < 8$

- (ix) Press F_1 as **Occupy**

Data will be stored in sensor

Uses of other functions keys as:-

F_7 : Satellite status: as example

Satellite No.	Orientation	Elevation	Azimuth
21	↑	72	27
18	↑	48	324
07	↓	38	110
15	↑	32	327
22	↑	38	298
06	↓	28	186

F_3 : Sky Status

F_4 : Health Status

F_8 : Stop & go Indication

F_9 : Position Status: as Example

Lat: $23^0 48' 58.89493' N$

Long: $86^0 26' 25.70666' E$

E Height: 195.008 m

F_2 : Cartesian coordinate as

$X = 362801.443m$

$Y = 5827169.694m$

$Z = 2559635.307m$

$HDOP = 1.4$

$VDOP = 2.2$

F_{10} : Battery and Memory Status

- (ix) Press F_1 : as stop
- (x) Press F_1 : as store
- (xi) Press ESC (To terminate the survey work)
- (xii) Press OK
- (xiii) Switch off From Terminal.

The above steps for both the sensors as Base Station and Rover Station. Firstly, the Base Station sensor at known Station should be switched on and follow up to step No. IX

Data Collection will be made in Rover sensor in following ways: -

- (i) **Static:** This process may be adopted to measure a longer baseline and time of observation should be more than 1 hour for better accuracy.
- (ii) **Rapid Static** This process may be adopted for base lines less than 20 km and observation time may be half an hour.
- (iii) **Kinematics:** This process may be adopted to get a path as Road, Railway line, River side etc. For observation time at first station for few minutes and move continuously on the path
- (iv) **Stop & Go Method:** This process may be adopted to know the surface features of the surveyed area. During observation move on the path continuously and to plot the features stay there for few seconds.

Finally switch off the Rover sensor first then the Base sensor.

Table: Satellite Status

Stn. No.	Satellite No.	Orientation	Azimuth	Bearing	Noise Ratio	Quality Index	Time	Remarks

EXPERIMENT: 10

Name of the Experiment:

Study of Terrestrial LIDAR (TLS) and demonstration for Topographical Surveying

Instruments:

1. 3D Laser Scanning Instrument
2. Tripod
3. Laser Scanning Prism with pole
4. 3m tape



Fig. 10.1 3D Laser Scanning Instrument

3D Laser Scanning:

Laser Scanning is an increasingly popular instrument for collecting vast amounts of accurate spatial data within a short amount of time. This makes laser scanners a popular piece of equipment within surveying.

Laser scanning is a popular surface surveying method that can accurately measure and collect data from objects, surfaces, buildings, and landscapes. Laser scanners collect information in the form of point cloud data, which consists of millions of 3D coordinates (XYZ coordinates).

Scanners collect millions of individual point measurements within minutes. The measurements are then plotted within a single XYZ coordinate system to form a 'point cloud' of the object's external surface. With the addition of GPS data, these points can be geo-referenced and transformed into a global reference system. Multiple data clouds collected from different viewpoints can also be combined ("registered") using common features to create one 3D dataset. The resulting 3D data can be used for visualization, modelling and planning in 3D, to use in BIM, geomorphological change and heritage projects. The point cloud data can also be

used to take any number of measurements without having to physically them on site. Point cloud data can form the basis of a rich dataset containing more than purely spatial information (XYZ). For example, intensity or photographic imagery could be applied to create a colored point cloud and 3D vector data can be over laid to map pipes and cables.

Principle of Laser Scanning:

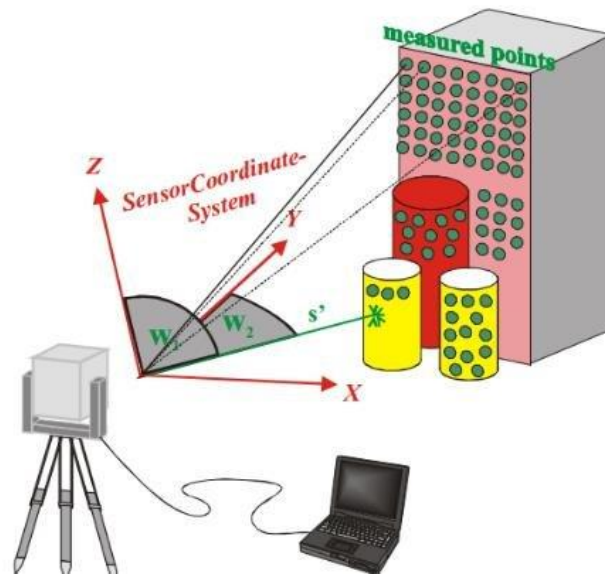


Fig. 10.2 3D Laser Scanning Principle

First, laser scanning systems throw out light waves that bounce off of surfaces and reflect back to the sensor. The sensor then calculates how far away the surface is by measuring the time taken for the light beam to complete its journey. This process is known as the “time of flight” measurement. The distance measured is then used to calculate a coordinate for the tiny section of the surface hit by the laser beam. All of this happens in just seconds, and during a single scan, a laser scanner will collect millions of 3D coordinates.

When the point clouds from laser scans are processed, they form a digital representation of the scanned surfaces, demonstrating the dimensions and spatial relationships of topographic features and structures.

Advantages of 3D Laser Scanning:

Laser scanning has a number of advantages over alternative surface surveying methods, such as photogrammetry. Here are some of the reasons why organizations use 3D scanning for surveys.

Laser Scanners Can Collect Data Quickly

Laser scanning is the speediest surface surveying method out there. Laser scanners can collect millions of data points in seconds, reducing the time and human resources needed to complete surveys. This method of surveying is perfect for projects that require a quick turnaround.

Systems Can Be Mounted to Vehicles

For projects that require information from vast landscapes or extensive networks of roads or railroads, laser scanning systems can be mounted onto land-based or aerial vehicles. The systems can then gather data from all of the necessary areas in a short space of time.

Laser Scanning Can Reduce Health and Safety Risks

Using laser scanners to collect information from dangerous or hard-to-reach locations means teams don't have to put themselves at risk. Instead, surveyors can stay safely out of harm's way while the scanner does its job.

3D Scanning Systems Can Survey Inaccessible Areas

Before laser scanning systems came about, certain areas were off-limits to surveyors. But with vehicle-mounted LiDAR systems, organizations can collect point cloud data from even the most inaccessible of areas. For example, drone-mounted LiDAR can survey unsafe sites and structures. Because laser scanning is a non-contact surveying method, it can also survey sensitive objects or areas, such as vulnerable historic sites.

Detailed 3D Models Can Be Created from Scan Data

Perhaps the biggest advantage of laser scanning is that scan data can create 3D CAD models. These 3D digital models can be used to analyze topographic features, structures, and transport corridors. They can also be used to simulate construction or urban planning projects.

Applications of Laser Scanning:

Laser scanning is utilized by many high-value, government-backed, and privately funded projects such as national defense and road surveying, Topographical Surveying, Open cast Mine Surveying and others surface surveying,. Some of its other well-known applications are listed below.

Spatial Scanning

Usually, the spatial scanning process is time-consuming, and one small mistake could affect the entire project. Laser scanners not only reduce the time taken in plotting and spatial scanning but also improve the planning process and considerably cut down the number of errors during the data collection. Because of the value laser scanners add to spatial scanning, researchers use them for mobile mapping, surveying, scanning buildings — and their interiors.

Road Surveying and Traffic Construction Analysis

Developing highways is one of the most challenging tasks for authorities as they can't stop traffic for too long to gather information about existing roads — but they can't delay

construction work either. Laser scanners help reduce turnaround time for road surveys and traffic construction analysis from weeks to a matter of hours.

Urban Topography

Urban topography is an interdisciplinary study of towns — their forms, layouts, architectural history, geography, etc. — for town planning. This is a critical process as it requires regular, accurate data collection by urban planners who use laser scanners to collect and process data before beginning development work.

Laser scanning can also be applied to civil surveying, reverse engineering, mining, and archaeological projects. It plays an important part in most civil engineering related projects and helps governments and private authorities accomplish tasks on time and with 100% accuracy.

How to Scanning the area for surveying

1. Switch on the instrument
2. Click on the Level Icon  & then done Centering & Levelling properly
3. Click on Cont.
4. Click on Manage
5. Click on Project
6. Click on New
7. Give a name of the project, description, etc & then press Enter 
8. Click on Store & click on Cont. when job file is selected
9. Click on back  & go to initial page
10. Click on Scanning
11. Click on Set Up when project is selected
12. Select Set up Method (Known Back sight) & select Ctrl pnt Project
13. Click on Station Id & click on New
14. Put the Station Id: . N: , E: , Ht: & click on Store
15. Put instrument Ht & click on Cont
16. Click on Back sight Id & click on New
17. Put the Station Id: . N: , E: , Ht: & click on Store
18. Put Target Ht
19. Click on Pick T (Pick method: Video Image)
20. Turn the instrument to bisect the B.S target properly
21. Select & then zoom the B.S target and then pick properly after clicking 
22. Click on  & click on Cont
23. Click on Set & then click on Yes

24. Click on Start (Task should be Scene & Image)

25. Click on back  after complete the scanning

How to Scanning the selected area for surveying

1. Complete up to the steep 23

2. Then click no Camera

3. Click on the icon just lick Stick 

4. Select upper left corner of the area to be surveyed by clicking & then click on 

5. Select lower right corner of the area to be surveyed by clicking & then click on 

6. Click on back  & then click on Start

7. Click on back  after complete the scanning